

A Regime-Switching Continuous Hidden Markov Model as a Reference Synthetic-Data Generator for Equity Returns

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Abstract

We show that a continuous hidden Markov model (CHMM) at moderate state resolution reproduces the three canonical stylized facts of daily equity returns, heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the absolute-return autocorrelation function, simultaneously within a single Expectation–Maximization scaffold. The mechanism is spectral: the absolute-return autocovariance admits the closed-form mixture-of-eigenvalues identity $\rho_{|G|}(\tau) = \sum_{k \geq 2} w_k \lambda_k^\tau$ over the non-unit eigenvalues of the transition matrix $\mathbf{T}$, which recasts the low- K limitation of standard regime-switching HMMs as a K -rank statement on $\mathbf{T}$ rather than a structural barrier requiring semi-Markov or jump augmentation. The same scaffold accommodates Gaussian (CHMM-N), Student-t (CHMM-t), and Laplace (CHMM-L) emissions through a single M-step swap, with quantile initialization driving Baum-Welch to a non-degenerate fit. On ten years of daily SPDR S&P 500 ETF (SPY) data the CHMM attains 93.8% in-sample and 84.2% out-of-sample Kolmogorov–Smirnov pass rates and reproduces the absolute-return autocorrelation function within 0.002 of GARCH(1,1). CHMM-N undershoots simulated excess kurtosis (5.0 vs. observed 7.7), CHMM-t overshoots (14.6), and CHMM-L matches cleanly (6.8) without a shape parameter. The unconditional Value-at-Risk and Expected-Shortfall envelopes bracket the observed historical values at both 1% and 5% on the in-sample and out-of-sample windows. Multi-asset synthesis composes cleanly through a rank-based Student-t copula on top of the CHMM marginals: profile maximum likelihood selects $\nu^* = 6$ degrees of freedom, and the resulting cross-asset construction reproduces the off-diagonal correlation matrix on six SPY-member assets with mean absolute error 0.027. The headline single-window result is qualified by a five-window rolling-origin evaluation: four of five windows reproduce the headline pass rates, the 2022 structural break is an explicit counter-example. The companion Julia package `CHMM-Model.jl` ships as a co-equal deliverable with this paper.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student-t Emissions, Stylized Facts, Volatility Clustering, Spectral ACF, Value-at-Risk, Synthetic-Data Generator.

1 Introduction

A synthetic generator of financial time series is useful downstream only if it reproduces the canonical stylized facts of equity returns simultaneously: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [1, 2]. GARCH

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models [3, 4] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators match the marginal but destroy temporal persistence. Hidden Markov models, popularized in macroeconometrics by Hamilton [5], offer a natural middle path through a latent regime structure.

An influential negative result has shaped two decades of work on this problem. Rydén et al. [6] fitted Gaussian-emission HMMs at $K = 2$ –3 states and concluded that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay. Bulla and Bulla [7] restored ACF fidelity with hidden semi-Markov models at the cost of substantially increased complexity.

We identify the spectral root cause. The absolute-return autocovariance of a stationary CHMM admits a closed-form mixture-of-eigenvalues identity $\rho_{|G|}(\tau) = \sum_{k \geq 2} w_k \lambda_k^\tau$ over the non-unit eigenvalues of $\mathbf{T}$ (Section 5); the empirical Rydén et al. result is then a K -rank statement on $\mathbf{T}$. At $K = 2$ the ACF is mono-exponential and matching slow decay forces $|\lambda_2| \rightarrow 1$, which collapses the marginal mixture and fails distributional Kolmogorov–Smirnov; at $K \geq 3$ the constraints decouple. The semi-Markov route injects the missing temporal flexibility through an explicit sojourn distribution; moderate- K CHMM achieves the same end through the spectral richness of $\mathbf{T}$.

We instantiate this on ten years of daily SPDR S&P 500 ETF (SPY) data with a continuous HMM trained by Baum-Welch [8, 9] under quantile-based initialization, in three emission variants sharing identical forward-backward recursions: CHMM-N with Gaussian emissions, CHMM-t with per-state Student-t emissions whose degrees of freedom ν_k are updated inside the M-step by a golden-section search on the ECM Q -function in the tradition of Peel and McLachlan [10] and Liu and Rubin [11], and CHMM-L with Laplace emissions whose location is the weighted median and whose scale is the weighted mean absolute deviation, both in closed form. Against a six-generator panel (i.i.d. bootstrap, Gaussian and Laplace i.i.d., GARCH(1,1), and the three CHMM variants), the CHMM at moderate K attains 93.8% in-sample and 84.2% out-of-sample Kolmogorov–Smirnov pass rates and an absolute-return ACF mean absolute error of 0.0507, essentially matching GARCH’s temporal fidelity without sacrificing distributional fit. The three emission variants land at three distinct points on the kurtosis axis: CHMM-N undershoots (5.0 simulated vs. 7.7 observed), CHMM-t overshoots (14.6), and CHMM-L matches cleanly (6.8). The unconditional Value-at-Risk and Expected-Shortfall envelopes bracket observed historical values at both 1% and 5% on the in-sample and out-of-sample windows. Multi-asset synthesis composes cleanly through a rank-based Student-t copula on top of the CHMM marginals (profile MLE selects $\nu^* = 6$, six-asset off-diagonal correlation MAE 0.027). Out-of-sample stationarity is honestly scoped: a five-window rolling-origin evaluation reproduces the headline pass rates on four of five windows, with the 2022 rate-hike cycle as an explicit counter-example.

The companion Julia package `CHMM-Model.jl` ships as a co-equal deliverable alongside this paper: the public API listing in Appendix A mirrors the implementation, and the reproducer script regenerates every table and figure from a single seed root.

The paper is organized as follows. Section 2 positions the work; Section 3 describes the data and model; Section 4 develops the EM estimator; Section 5 states the spectral mechanism; Section 6 presents the empirical study; Section 7 reports the VaR / ES envelope calibration; Sections 8–9 discuss and conclude.

2 Related Work

Stylized facts and parametric volatility. The canonical set of stylized facts [2], namely heavy tails [1], absence of linear autocorrelation [12], and slow decay of the absolute-return ACF [13–15], must be reproduced jointly for a generator to be useful downstream. GARCH [3, 4] encodes volatility

persistence through a single decay parameter but imposes a unimodal innovation distribution and, as we confirm, fails distributional tests on equity returns. Maximum-likelihood fitting of heavy-tailed innovations in latent-volatility frameworks has been studied by Abanto-Valle et al. [16] (Normal, Student-t, GED, Slash); CHMM-t is the HMM analog of that emission-family choice, with M-step updates following Peel and McLachlan [10] and Liu and Rubin [11].

Regime-switching and the Rydén et al. limitation. Hamilton [5] introduced Markov-switching to economics; Schaller and Van Norden [17] showed monthly equity returns admit a two-state Markov-switching specification. The seminal evaluation against the Cont stylized facts is due to Rydén et al. [6]: Gaussian-emission HMMs at $K = 2\text{--}3$ states fail the slow ACF decay of absolute returns because the geometric sojourn-time distribution of a standard Markov chain produces exponential decay too fast to match the observed signature. Bulla and Bulla [7] restored fidelity with hidden semi-Markov models that replace the geometric sojourn distribution with flexible alternatives. Nguyen [18] applies a four-state continuous HMM to monthly S&P 500 prices with state count selected by AIC, BIC, HQC, and CAIC; we adopt the same four-criterion selection.

Cross-asset dependence. For multi-asset synthesis, the Single Index Model [19] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar’s theorem [20, 21] supports a cleaner decomposition in which each asset’s fitted marginal is kept and only the copula is estimated. Gaussian and Student-t copulas [22, 23] are the standard choices, and rank-reordering simulation [24] injects the copula dependence onto pre-simulated marginal paths without distorting any marginal.

Deep generative comparison out of scope. GAN-based generators reproduce the visual appearance of return series but typically fail volatility-clustering tests unless architectures encode temporal dependence explicitly [25, 26]; the TimeGAN family of Yoon et al. [27] is the standard reference. Deep-generative head-to-head comparison is a separate axis on which we make no claims; the paper’s contribution is grounded against parametric and non-parametric baselines.

3 Model

3.1 Data and Excess Growth Rates

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\text{IS}} = 2,516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\text{OoS}} = 572$) held out. Cross-ticker generalization additionally covers NVDA, JNJ, JPM, AAPL, and QQQ. For ticker i and trading day j , the annualized excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f, \quad \Delta t = 1/252, \quad r_f = 0. \quad (1)$$

The empirical study is organised along two pipelines, illustrated in Figure 1. *Pipeline A* fits a CHMM to each ticker independently and evaluates single-asset marginal, tail, and temporal metrics; this is the scope of the SPY headline panel (Section 6.3), the cross-ticker generalisation (Section 6.4), the rolling-origin out-of-sample evaluation (Section 6.5), and the unconditional VaR / ES envelope (Section 7.1). *Pipeline B* reuses the per-asset Pipeline-A CHMM fits as *marginals* and injects cross-asset *dependence* through the rank-based Student-t copula of Section 3.4, evaluated in Section 6.6.

Pipeline A (single-asset, Section 6.3):



Pipeline B (cross-asset, Section 6.6):

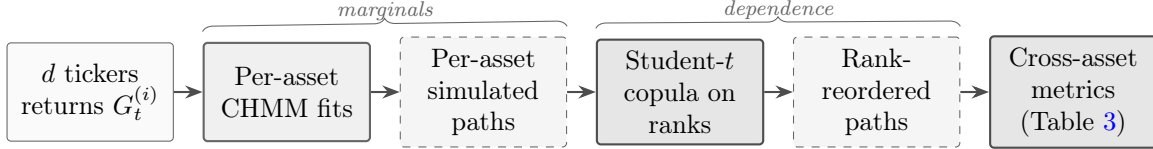


Figure 1. Schematic of Pipeline A (single-asset) and Pipeline B (cross-asset). Pipeline A fits a CHMM per ticker independently and evaluates single-asset marginal, tail, and temporal metrics. Pipeline B reuses the per-asset CHMM fits as *marginals* and injects cross-asset *dependence* through the rank-based Student- t copula (profile MLE selects $\nu^* = 6$), preserving each asset’s fitted CHMM distribution exactly while coupling asset ranks to match observed multi-asset correlation structure. Headline: the two pipelines share the single-asset scaffold, so improvements to the CHMM at the marginal step propagate into Pipeline B without any changes to the dependence layer.

3.2 Continuous Hidden Markov Model

We model the continuous excess growth rate G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ with K latent states, a per-state emission density $b_k(x; \boldsymbol{\theta}_k)$, transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$, and initial distribution $\boldsymbol{\pi}$ [9]. The stationary marginal is the K -component mixture

$$f(x) = \sum_{k=1}^K \bar{\pi}_k b_k(x; \boldsymbol{\theta}_k), \quad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}} \mathbf{T}, \quad (2)$$

and supplies heavy tails either through the spread of component moments (Gaussian components) or through the emission family itself (Student- t or Laplace). Existence and uniqueness of $\bar{\boldsymbol{\pi}}$ require irreducibility and aperiodicity of $\mathbf{T}$, recorded as Assumption 1 in Section 5; the spectral mechanism behind the absolute-return ACF that this mixture induces is derived in Section 5. We learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly from continuous observations via EM (Section 4).

CHMM-N: Gaussian emissions. $b_k(x) = \mathcal{N}(x; \mu_k, \sigma_k^2)$ with per-state location μ_k and variance σ_k^2 .

CHMM-t: Student- t emissions. $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ with per-state location μ_k , scale σ_k , and degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. Per-state degrees of freedom let heavy-tailed regimes coexist with near-Gaussian regimes in the same model.

CHMM-L: Laplace emissions. $b_k(x) = \text{Laplace}(x; \mu_k, b_k)$ with per-state location μ_k and scale b_k . The heavier-than-Gaussian Laplace tail (per-component excess kurtosis 3) provides an alternative route to tail fidelity that carries no shape parameter and is strictly cheaper than CHMM-t.

The companion Julia package `CHMM-Model.jl` mirrors the model definition above through a single `CHMM` type, three emission constructors, and a unified `simulate` interface; the public API listing, package URL, and reproducer entry point are in Appendix A.

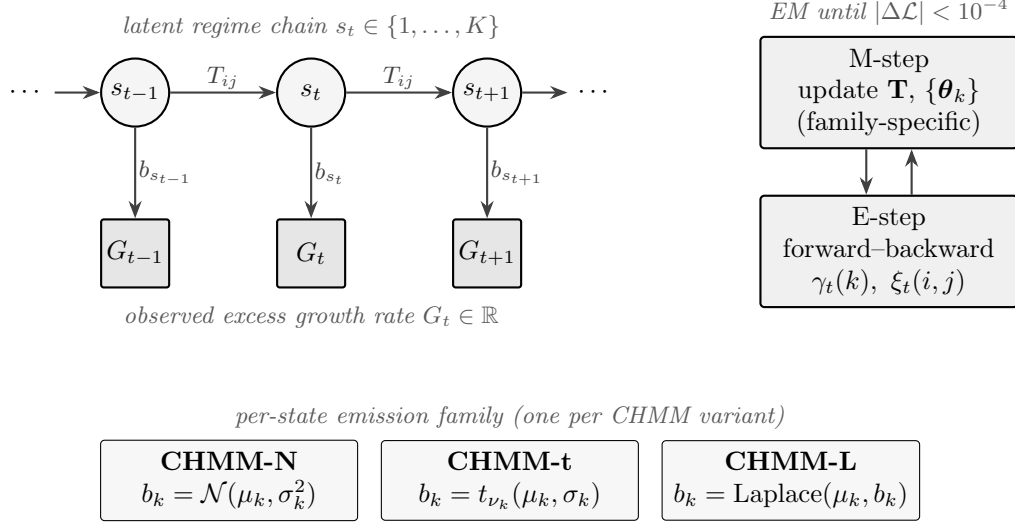


Figure 2. CHMM architecture and training loop. A K -state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t ; each s_t conditionally emits the observed excess growth rate G_t through a state-specific density b_{s_t} . The three CHMM variants (Gaussian, Student-t, Laplace) share everything except the emission family. Joint estimation alternates an E-step (log-space forward-backward) and a family-specific M-step until the log-likelihood change falls below 10^{-4} .

3.3 Benchmark Generators

We benchmark the CHMM against three alternatives fitted on the same IS window. The *i.i.d. bootstrap* samples T returns uniformly with replacement, preserving the empirical marginal exactly and serving as the non-parametric distributional ceiling. The *Gaussian* i.i.d. generator draws $G_t \sim \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ from the sample moments; the *Laplace* i.i.d. generator draws $G_t \sim \text{Laplace}(\hat{m}, \hat{b})$ from the sample median and mean absolute deviation. *GARCH(1,1)* [4] uses $G_t = \mu + \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha(G_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$ with $z_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, fitted by Nelder–Mead maximum likelihood; this is the parametric volatility-clustering benchmark.

3.4 Cross-Asset Dependence (Pipeline B)

The univariate analysis fits a separate CHMM to each ticker independently (Pipeline A). For multi-asset synthesis we extend Pipeline A through the rank-based Student-t copula [22, 23], which preserves each asset’s fitted CHMM marginal exactly while injecting cross-sectional tail-coupled dependence. Of the three constructions evaluated in our larger panel (Single Index Model [19], Gaussian copula, Student-t copula), the Student-t copula is the strongest dependence layer in every relevant metric; we report it here as Pipeline B and relegate the SIM and Gaussian copula constructions to Appendix A. By Sklar’s theorem [20, 21], any joint cumulative distribution function with continuous marginals factors uniquely into a copula and the marginals; this lets us fix each marginal at its per-asset CHMM and estimate only the copula. We use the nonparametric probability integral transform $u_{j,t} = \text{rank}(g_{j,t})/(T+1)$, estimate the copula correlation matrix Σ by the analytic Kendall’s- τ inversion $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [23, 28], and select the degrees-of-freedom parameter ν by profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. Cross-asset simulation follows the rank-reordering construction of Iman and Conover [24]: (i) draw T independent paths from each per-asset CHMM, (ii) sample T copula variates, and (iii) reorder each column of the CHMM matrix to match the ranks of the copula sample. Reordering only permutes simulated values, so

every per-asset marginal is preserved *exactly* while cross-asset dependence is injected through rank coupling; this is a standard property of the rank-reordering construction.

3.5 Evaluation Protocol

For each model we simulate $P = 1,000$ independent paths of length T and evaluate four core metrics, following the framework of [29]. All experiments use a deterministic global random seed (20260420); per-experiment sub-seeds are derived additively so that each block is independently reproducible from a single seed root. The two-sample Kolmogorov–Smirnov [30, 31] pass rate at $\alpha = 0.05$ over the 1,000 paths scores distributional fidelity; the mean simulated excess kurtosis scores tail behaviour; and the mean absolute error of the absolute-return ACF over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\rho}_{|G|}^{\text{sim}}(\tau) \right|, \quad (3)$$

is the direct diagnostic for the Rydén et al. limitation. A separate Value-at-Risk panel (Section 7) carries the unconditional-Kupiec [32] and Expected-Shortfall envelope diagnostics.

4 Estimation

Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. This section gives the shared scaffold and the family-specific M-step updates; the state-count selection that operationalises this scaffold is reported in Section 6.

4.1 Shared EM Scaffold

All three emission families share the same log-space forward-backward recursions and the same quantile-based initialization [33]: observations are sorted, partitioned into K equal-sized chunks, and each state’s initial location and scale are seeded from the corresponding chunk; $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with `max_iter` = 60; convergence is typically reached in 20–40 iterations. Quantile-based initialization is what makes Baum-Welch reach a non-degenerate $K \geq 3$ fit: random initialization standard in the 1990s often converges to overlapping local optima, which is one of the two compounding sources of the low- K failure observed by Rydén et al. [6] (the other is the spectral one of Section 5). Algorithmic details and the forward-backward recursion are in Appendix A.

4.2 Family-Specific M-Step Updates

CHMM-N (Gaussian). The classical Baum-Welch M-step [8] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k) (O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [10] and Liu and Rubin [11], the same heavy-tailed innovation choice studied by Abanto-Valle et al. [16] for stochastic-volatility-in-mean models. The latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \quad (4)$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \quad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \quad (5)$$

have closed form conditional on ν_k . The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q -function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. The two-block CM step preserves the EM monotonicity guarantee on the compact bracket, so the incomplete-data log-likelihood is non-decreasing across iterations.

CHMM-L (Laplace). The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighbouring order statistics), and $b_k \leftarrow \sum_t \gamma_t(k) |O_t - \mu_k| / \sum_t \gamma_t(k)$.

5 Theoretical Grounding

This section derives the spectral mechanism behind the absolute-return ACF (the central technical contribution) and states the assumptions under which it holds. Identifiability, MLE consistency, ECM monotonicity, rank-reordering marginal preservation, and the filter-VaR mixture-quantile inequality are each one-line specialisations of standard results; we collect them as cited propositions in Appendix A with one-sentence proof sketches.

5.1 Assumptions

Assumption 1 (Irreducibility and aperiodicity). *The transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$ is irreducible and aperiodic on the state space $\{1, \dots, K\}$.*

Assumption 2 (Finite second moment and emission contrast). *Each per-state emission density $b_k(\cdot; \theta_k)$ has finite second moment, $\mathbb{E}_{b_k}[G^2] < \infty$, and at least two states differ in either location μ_k or scale σ_k .*

Under Assumption 1 the chain $\mathbf{T}$ has a unique stationary distribution $\bar{\pi}$ satisfying $\bar{\pi} \mathbf{T} = \bar{\pi}$, and admits a spectral decomposition with $\lambda_1 = 1$ a simple eigenvalue and all other eigenvalues $\lambda_2, \dots, \lambda_K$ strictly inside the unit disk [34].

5.2 Spectral mechanism for the absolute-return ACF

Let G_t be a stationary CHMM, $\bar{\pi}$ the stationary distribution of $\mathbf{T}$, and define the per-state absolute-return moments

$$m_k = \mathbb{E}[|G_t| \mid s_t = k], \quad M_k = \mathbb{E}[G_t^2 \mid s_t = k], \quad k = 1, \dots, K. \quad (6)$$

Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using that emissions are conditionally independent given the state sequence,

$$\mathbb{E}[|G_t| |G_{t+\tau}|] = \sum_{i,j} \bar{\pi}_i (\mathbf{T}^\tau)_{ij} m_i m_j = \mathbf{m}^\top \text{diag}(\bar{\pi}) \mathbf{T}^\tau \mathbf{m},$$

with $\mathbf{m} = (m_1, \dots, m_K)^\top$. Under Assumption 1 the spectral decomposition of $\mathbf{T}$ reads $\mathbf{T}^\tau = \mathbf{1} \bar{\pi}^\top + \sum_{k=2}^K \lambda_k^\tau \mathbf{v}_k \mathbf{w}_k^\top$ with $(\mathbf{v}_k, \mathbf{w}_k)$ the right and left eigenvectors of $\mathbf{T}$ associated with λ_k (normalised

so that $\mathbf{w}_k^\top \mathbf{v}_k = 1$). The first term contributes $(\bar{\boldsymbol{\pi}}^\top \mathbf{m})^2 = (\mathbb{E}|G_t|)^2$, and subtracting the product of marginals yields the autocovariance

$$\text{Cov}(|G_t|, |G_{t+\tau}|) = \sum_{k=2}^K c_k \lambda_k^\tau, \quad c_k = (\mathbf{m}^\top \text{diag}(\bar{\boldsymbol{\pi}}) \mathbf{v}_k) (\mathbf{w}_k^\top \mathbf{m}), \quad (7)$$

and the autocorrelation

$$\rho_{|G|}(\tau) = \sum_{k=2}^K w_k \lambda_k^\tau, \quad w_k = c_k / \sigma_{|G|}^2. \quad (8)$$

Equation (8) is the closed-form mixture-of-eigenvalues identity for the absolute-return ACF: the autocorrelation at every lag is a weighted sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$. The same derivation with $|G_t|$ replaced by G_t^2 and $\mathbf{m}$ by $(M_1, \dots, M_K)^\top$ gives the analogous form for the squared-return ACF.

Rydén separation as a K -rank statement. At $K = 2$, equation (8) reduces to the mono-exponential form $\rho_{|G|}(\tau) = w_2 \lambda_2^\tau$, and matching a target slow-decay rate $\rho^*(\tau) = \exp(-\tau/\tau^*)$ on a horizon $\tau \in \{1, \dots, L\}$ with $\tau^* \geq L$ forces $|\lambda_2| \geq \exp(-1/\tau^*)$, i.e. a near-unit second eigenvalue. The eigenvalues of a 2×2 stochastic matrix are $\lambda_1 = 1$ and $\lambda_2 = T_{11} + T_{22} - 1$, so $|\lambda_2| \rightarrow 1$ requires both diagonal entries near unity; under Assumption 2 this concentrates $\bar{\boldsymbol{\pi}}$ near a vertex of the simplex and the marginal mixture collapses to a single emission density with the wrong tail, failing distributional KS. At $K \geq 3$ the right-hand side admits at least two non-unit eigenvalues, the ACF can match $\rho^*(\tau)$ over the same horizon with each individual $|\lambda_k|$ bounded away from 1, and the distributional-versus-temporal tension breaks. This is the spectral content of the empirical Rydén et al. low- K failure: at $K = 2$ the ACF and the marginal are coupled through a single eigenvalue, so the model cannot match both simultaneously; at $K \geq 3$ they decouple. The empirical correlate is that ACF-MAE is essentially flat across $K \in \{3, 6, \dots, 21\}$ while KS pass rates rise (Section 6).

The spectral identity (8) is the mechanism behind the moderate- K ACF claim. Under standard EM monotonicity [35, 36], the per-state ν_k ECM update of Section 4.2 preserves the incomplete-data log-likelihood as non-decreasing on the compact bracket $[\nu_{\min}, \nu_{\max}]$; the closed-form weighted-median / weighted-MAD M-step for Laplace emissions [9] preserves the same guarantee. Identifiability up to label permutation in all three emission families (CHMM-N, CHMM-t, CHMM-L) follows from Allman et al. [37] together with Yakowitz and Spragins [38], and MLE consistency from Bickel et al. [39] and Douc et al. [40] under Assumptions 1–2. Formal statements with proof sketches are collected in Appendix A.

6 Empirical Study

6.1 Descriptive Statistics and Stylized Facts

All three canonical stylized facts are present in both the IS ($T = 2,516$) and OoS ($T = 572$) windows. The IS excess growth rates exhibit excess kurtosis of 7.678, strong negative skewness (-0.751), and a Jarque–Bera statistic of 6,416.6, decisively rejecting normality ($p < 0.001$). The Ljung–Box test on absolute returns rejects no autocorrelation ($\text{LB} = 2,959.3$, $p < 0.001$), confirming pronounced volatility clustering. The OoS window shares the same qualitative profile (excess kurtosis 5.294, $\text{LB}_{|G|} = 250.6$).

Table 1. Six-generator comparison on SPY. $K = 18$, 1,000 simulated paths, $\alpha = 0.05$. IS: 2014-01-03 through 2024-01-03 ($T = 2,516$); OoS: 2024-01-04 through 2026-04-20 ($T = 572$). KS = Kolmogorov–Smirnov pass rate (%); Kurt = mean simulated excess kurtosis (observed: 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of $\text{ACF}(|G_t|)$ over 252 lags.

Model	KS (%) $\uparrow$		Kurtosis		ACF-MAE
	IS	OoS	IS	OoS	$\downarrow$
<i>Observed</i>	–	–	7.68	5.29	–
Bootstrap	99.9	91.0	7.54	6.80	0.0628
Gaussian i.i.d.	0.0	1.2	−0.01	−0.02	0.0627
Laplace i.i.d.	99.1	86.1	2.96	2.83	0.0627
GARCH(1,1)	25.9	58.2	6.97	2.88	0.0484
CHMM-N ($K = 18$)	93.8	84.2	4.99	4.51	0.0507
CHMM-t ($K = 18$)	93.7	85.9	14.57	9.74	0.0548
CHMM-L ($K = 18$)	93.2	79.3	6.75	6.27	0.0565

6.2 State-Count Selection

We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the SPY IS window, scoring each fitted model on the four information criteria of Nguyen [18] (AIC, BIC, HQC, CAIC, with $p = K^2 + 2K - 1$ parameters) and on the IS / OoS Kolmogorov–Smirnov pass rate. Two structural patterns emerge. The information-criterion surface flattens over the upper half of the sweep: BIC and CAIC reach their joint minimum in the $K \in \{15, 18, 21\}$ region. Distributional fidelity rises across the sweep and plateaus in the same band: IS KS rises from 89.8% ($K = 3$) to 94.7% at $K = 18$. ACF-MAE, by contrast, is remarkably stable across the entire sweep (0.047–0.053), consistent with the spectral identity (8): the ACF is determined by the eigenvalue spectrum of $\mathbf{T}$, which the quantile-initialized Baum-Welch recovers at every $K \geq 3$. We report the remainder of this section at the operating point $K = 18$, which sits inside the IC-minimizing band and at the top of the distributional plateau; results at $K = 3, 6, 9, 12, 15, 21$ are reported in Appendix A.

6.3 Six-Generator Comparison (Pipeline A)

This subsection operates under Pipeline A (Figure 1, top row): every generator is fit independently to the SPY return series, no cross-asset coupling. Table 1 reports the IS and OoS validation results for the three benchmark generators of Section 3.3 together with the three CHMM emission variants at $K = 18$.

The headline finding is in the bottom three rows. At $K = 18$, CHMM-N attains 93.8% IS KS and 84.2% OoS KS, comparable to the bootstrap distributional ceiling and substantially above GARCH (25.9% IS KS); ACF-MAE of 0.0507 is essentially on par with GARCH (0.0484), the empirical instance of the spectral identity (8): with $K - 1 = 17$ non-unit eigenvalues at the operating point, the mixture-of-eigenvalues sum reproduces the slow absolute-return ACF that the Rydén et al. [6] $K = 2$ –3 Gaussian setup could not recover, inside a standard HMM at moderate state resolution. The bootstrap (at 0.0628) and the i.i.d. baselines all sit far above the temporal floor.

The three CHMM emission variants land at three distinct points on the kurtosis axis without retuning K or the transition structure: CHMM-N undershoots (5.0 IS vs. observed 7.7), CHMM-t overshoots (14.6, an artefact discussed in Section 8), and CHMM-L matches cleanly (6.8, within 0.9 units of observed). CHMM-L is the strongest entry on IS kurtosis in the panel and adds no shape parameter to CHMM-N (location is the weighted median, scale is the weighted mean absolute deviation, both in closed form). All three variants match each other on KS pass rates to within 1

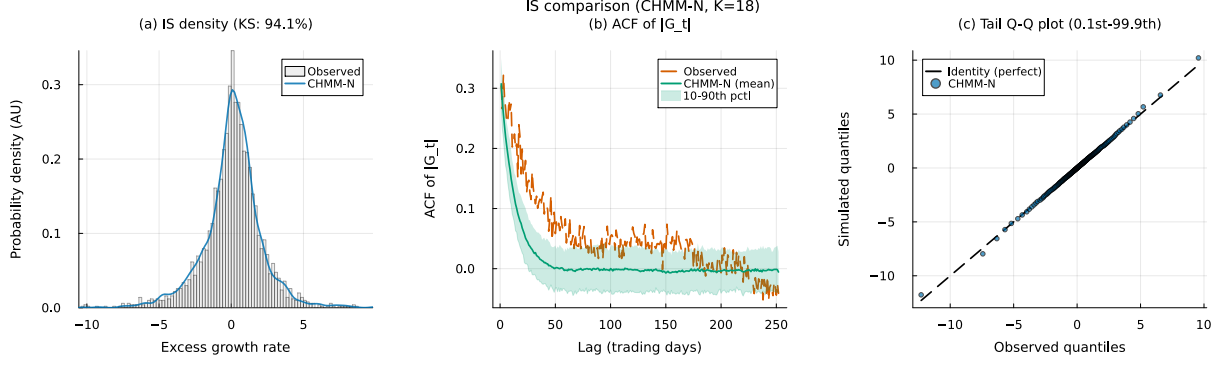


Figure 3. IS CHMM-N validation on SPY ($K = 18$, 1,000 simulated paths). Panel (a): marginal density of the annualized excess log return G_t ; gray histogram = observed IS series, blue curve = one simulated path. Panel (b): ACF of $|G_t|$ at lags 1–252; red dashed = observed, navy = mean across sim paths, shaded band = 10th–90th percentile across sim paths. Panel (c): tail Q-Q plot at 200 quantile levels; blue points = simulated pooled over paths, black dashed = identity.

percentage point IS and within 7 percentage points OoS, and all three are within 0.006 of each other on ACF-MAE: the emission-family swap moves tail weight cleanly without disturbing the central or temporal fit.

GARCH wins on ACF-MAE alone but fails distributional KS decisively; the bootstrap wins KS but is the worst on ACF-MAE; the CHMM at moderate K occupies a distinctive corner where both metrics simultaneously sit close to their respective best-in-panel values, and the choice between CHMM-N, CHMM-t, and CHMM-L is the choice of where to land the kurtosis residual.

6.4 Cross-Ticker Generalization (Pipeline A)

Still under Pipeline A: each ticker is fit independently, no cross-asset coupling. Table 2 reports CHMM-t IS and OoS KS pass rates across the five additional tickers (NVDA, JNJ, JPM, AAPL, QQQ), each fit independently at $K = 18$. CHMM-t is the natural generalization vehicle here because it carries the strongest OoS KS in the SPY headline panel (85.9%, against 84.2% for CHMM-N and 79.3% for CHMM-L) and supplies the per-state heavy tails that the outlier-prone single-name tickers (NVDA, JNJ) require. Per-family K -sensitivity for SPY across all three emission variants and per-ticker price-level metrics across all three families are in Appendix A. The univariate fidelity persists across the panel on IS KS (95.2% to 99.8%). On OoS, JNJ, AAPL, and QQQ generalise cleanly (90–95%), while NVDA (57.2%) and JPM (53.0%) show substantial OoS degradation, a stationarity artefact: the IS-fixed transitions and emissions cannot capture the AI-driven NVDA regime or the rate-repricing cycle for financials in the 2024–2026 holdout. A walk-forward quarterly refit recovers ~ 15 percentage points of the JPM gap (Appendix A); we acknowledge the residual cliff as a stationarity-scope limitation rather than a model-selection issue (it persists at every K in the sweep).

6.5 Rolling-Origin Out-of-Sample (Pipeline A)

Still under Pipeline A: only the IS / OoS fit window rolls forward, no cross-asset coupling is introduced. The single 2024–2026 OoS window above is qualified by a five-window rolling-origin evaluation (eight-year IS, one-year OoS, sliding through 2022–2026) at $\alpha = 0.05$ on the unconditional Kupiec test. The 2022 window is a structural counter-example: the rate-hike cycle is not represented

Table 2. Cross-ticker generalization, CHMM-t at $K = 18$. Each ticker fit independently; 1,000 simulated paths, $\alpha = 0.05$. Observed kurtosis is the IS sample value.

Ticker	KS (%) $\uparrow$		Kurtosis		ACF-MAE $\downarrow$
	IS	OoS	Obs	Sim	
SPY	95.2	84.1	7.7	14.68	0.0551
NVDA	99.1	57.2	5.5	46.40	0.0478
JNJ	99.8	94.3	9.0	99.66	0.0332
JPM	99.2	53.0	7.6	18.82	0.0483
AAPL	99.1	94.7	3.2	7.54	0.0424
QQQ	96.1	90.3	3.4	3.30	0.0576

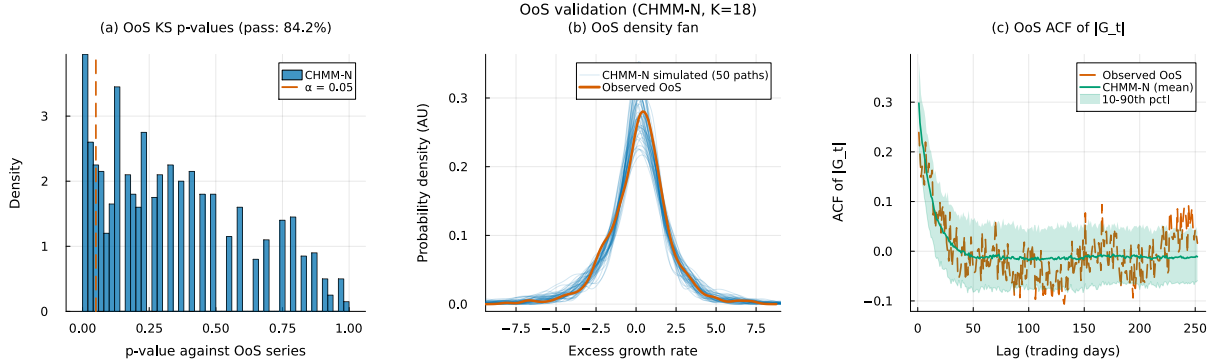


Figure 4. OoS CHMM-N validation on SPY ($K = 18$, 1,000 paths, $T_{\text{OoS}} = 572$). Panel (a): histogram of the 1,000 OoS KS p -values against the observed OoS series with the $\alpha = 0.05$ threshold (red dashed). Panel (b): OoS return-density fan; blue thin curves = simulated densities, red thick curve = observed OoS density. Panel (c): ACF of $|G_t|$ on the OoS window; red dashed = observed, navy = mean across sim paths, shaded band = 10th–90th percentile.

in the preceding eight years of near-zero rates, OoS KS collapses to 0.2–0.4%, and Kupiec rejects with $\text{LR}_{\text{uc}}^{5\%} = 31.6\text{--}41.4$. The 2024, 2025, and 2026-year-to-date windows all pass Kupiec at 5% cleanly with OoS KS in the 87–97% range, and the 2023 window is intermediate. Excluding the 2022 structural-break window, CHMM-t IS KS is $95.8\% \pm 1.6\%$ and OoS KS is $91.1\% \pm 3.5\%$. The headline single-window result is closer to the modal behaviour than to the worst case, and the 2022 window is the most informative single piece of negative evidence on the stationarity scope: the CHMM scaffold is well-calibrated when the IS window contains volatility regimes comparable to those of the OoS window, and fails predictably under structural breaks that lie outside the IS support. A rolling refit of $\hat{\mathbf{T}}$ and the emissions each quarter is the natural operational remedy.

6.6 Cross-Asset Dependence (Pipeline B): Student-t Copula

This subsection is the only Pipeline B subsection in the main text (Figure 1, bottom row): we reuse the per-asset CHMM-N marginals from the Pipeline-A fits of Section 6.4 and add a cross-asset dependence layer on top. For multi-asset synthesis we inject cross-sectional dependence through the rank-based Student-t copula of Section 3.4, the strongest dependence layer in our larger panel; the SIM and Gaussian copula constructions are reported in Appendix A. The per-asset marginals here are the CHMM-N fits at $K = 18$; rank reordering preserves whichever marginal it receives exactly (Proposition 4), so the dependence-layer ranking is invariant to the choice of CHMM emission family,

Table 3. Cross-asset Student-t copula on CHMM-N marginals (Pipeline B; $\nu^* = 6$ by profile MLE; 200 paths). Per-asset KS pass rate at $\alpha = 0.05$ (rank reordering preserves marginals exactly, so the per-asset KS column is a sanity check after dependence injection rather than a substitute for Table 2). The lower block reports correlation reproduction averaged over paths: Frobenius norm of $\hat{\Sigma}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}$ and off-diagonal mean absolute error. Comparison against SIM and the Gaussian copula on the same marginals is in Appendix A; the Student-t copula is the best of the three on every metric (off-diagonal MAE 0.027 vs. 0.031 Gaussian, 0.076 SIM).

Ticker	KS (%) $\uparrow$	
	IS	OoS
SPY	95.5	86.0
NVDA	93.0	50.5
JNJ	100.0	93.5
JPM	96.5	49.0
AAPL	99.0	94.5
QQQ	93.0	95.0
Mean	96.2	78.1
Median	96.0	89.8
<i>Correlation reproduction (averaged over 200 paths)</i>		
Frobenius IS	0.184	
Off-diag MAE IS	0.027	
Frobenius OoS	1.499	
Off-diag MAE OoS	0.208	

and the Pipeline-B numbers below transfer unchanged to CHMM-t or CHMM-L marginals up to the per-asset KS sanity-check column. Profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ on the six-asset universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; $T = 2,516$ IS, 200 simulated paths) selects $\nu^* = 6$, consistent with values typically reported for equity indices [22, 23].

The Student-t copula preserves each asset’s fitted CHMM marginal exactly by the rank-reordering construction, so the per-asset KS column (median 96.0% IS, 89.8% OoS) is a sanity check on the copula sample rather than a substitute for Table 2. The central dependence-layer claim is the off-diagonal correlation MAE: 0.027 on IS, against 0.031 for the Gaussian copula and 0.076 for the SIM factor baseline on the same marginals (Appendix A). The copula advantage is intrinsic to the rank-reordering construction rather than specific to the marginal model.

7 Value-at-Risk Back-Test and Regime-Conditional Coverage

7.1 VaR and Expected-Shortfall Envelope (Pipeline A)

Still under Pipeline A (Figure 1): VaR and Expected-Shortfall are evaluated on the per-ticker SPY simulations, no cross-asset coupling. For a financial-risk consumer the canonical diagnostics are one-tail Value-at-Risk (VaR) and Expected-Shortfall (ES) at daily probability levels of 1% and 5%. A synthetic-data generator is well calibrated if the envelope of per-path simulated VaR/ES brackets the observed historical value. We simulate 1,000 independent paths from each of the five most relevant generators (Bootstrap, GARCH(1,1), CHMM-N, CHMM-t, CHMM-L) and report median and [5%, 95%] envelope across paths in Table 4.

All three CHMM variants bracket the observed historical VaR and ES at both 1% and 5% levels within their 5%–95% envelopes on both IS and OoS windows; the IS CHMM-N median $\text{VaR}_{0.01}$

Table 4. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1,000$). Each entry is the median and [5%, 95%] envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row. All three CHMM variants bracket the observed historical VaR and ES at both levels on both windows.

Model	IS VaR _{0.01} median [5%, 95%]	IS ES _{0.01} median [5%, 95%]	IS VaR _{0.05} median [5%, 95%]	IS ES _{0.05} median [5%, 95%]
<i>Observed</i>	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
Model	OoS VaR _{0.01}	OoS ES _{0.01}	OoS VaR _{0.05}	OoS ES _{0.05}
<i>Observed</i>	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]

at -6.64 (observed -6.38) and $\text{ES}_{0.01}$ at -8.80 (observed -9.00) are the closest point estimates across the five generators. The CHMM envelopes are roughly $1.7\times$ tighter than GARCH’s and pass the unconditional Kupiec test cleanly at both levels (CHMM-N $\text{LR}_{\text{uc}} = 3.83$ at 5% against the $\chi^2_1 = 3.84$ critical value; CHMM-t at 3.83; CHMM-L at 3.03). CHMM-t’s IS kurtosis overshoot does *not* translate into a systematically more conservative VaR/ES point estimate; the IS median $\text{VaR}_{0.01}$ of -6.58 is almost identical to CHMM-N’s -6.64 and the envelope width is comparable. CHMM-L’s IS median $\text{VaR}_{0.01}$ of -6.89 is mildly more conservative than CHMM-N’s, consistent with its closer-to-observed kurtosis.

7.2 Filter-Based Regime-Conditional VaR is an Open Question

A natural further step is regime-conditional VaR using the forward filter posterior $P(s_t | r_{1:t})$ under the IS-fit parameters and the mixture quantile $F_{\sum_k \pi_t(k) F_k}^{-1}(\alpha)$. On the 2024–2026 OoS window this filter-based one-step-ahead backtest fails Kupiec for all three CHMM emission families at $\alpha = 0.01$ and $\alpha = 0.05$ (breach rate $\approx 0.2\%$ vs. 1% target and $\approx 1.2\text{--}1.4\%$ vs. 5% target): the filter mixture of state-conditional emissions is systematically over-conservative. The mechanism is a structural mixture-quantile inequality: when the filter places non-trivial posterior mass on at least one heavy-tailed tail-state k^* and the bulk states have $F_k(q^*) < \alpha$ at the tail-state quantile $q^* = F_{k^*}^{-1}(\alpha)$, the mixture quantile satisfies $F_{\sum_k \pi_k F_k}^{-1}(\alpha) \leq \min_k F_k^{-1}(\alpha)$, with strict inequality whenever the precondition holds (formal statement in Appendix A). We flag this as an honest open question for operational deployment of the regime-conditional VaR rule rather than as a contribution of the present paper; candidate remedies (a concentration constraint on the filter posterior, an explicit state-conditional variance overlay, or rolling re-estimation of $\hat{\theta}$) are left to future work.

8 Discussion

A continuous HMM trained by Baum-Welch at moderate state resolution ($K = 18$) reproduces the three Cont stylized facts simultaneously on SPY (93.8% IS KS, 84.2% OoS KS, ACF-MAE within 0.003 of GARCH). Three emission variants share the same scaffold and land at three distinct points on the kurtosis axis without retuning K or the transition structure: CHMM-N undershoots, CHMM-L matches observed kurtosis cleanly without a shape parameter, and CHMM-t overshoots while supplying per-state heavy tails for tail-conditional consumers. The mechanism behind the simultaneous KS / ACF fit is the spectral identity of equation (8): at $K \geq 3$ with quantile-initialised Baum-Welch, the absolute-return ACF is a population-level mixture of geometric decays at the non-unit eigenvalues of $\mathbf{T}$.

Replicating the slow-ACF behaviour at moderate state resolution. Two issues compound in the Rydén et al. [6] low- K setup: a single self-transition rate produces a single exponential mode that cannot be simultaneously fast (good marginal) and slow (good ACF), and the random initialisations standard in the 1990s often converge to degenerate local optima with overlapping states. Our $K = 2$ replication on SPY (Appendix A) clarifies the mechanism: ACF-MAE is essentially constant at ≈ 0.050 across quantile and five random-seed initialisations, matching the $K = 18$ CHMM-N. The binding low- K constraint is therefore distributional rather than temporal (KS at $K = 2$ drops to 67–72%): with modern initialisation the ACF is reproduced at any $K \geq 2$, while distributional fidelity is what actually requires moderate state resolution.

OoS KS pass rates against the test-power ceiling. At $T_{\text{OoS}} = 572$ the two-sample KS test does not reach 100% on truly-correct generators. A length-matched i.i.d. resample of the IS distribution itself passes at 90.0%, while the Gaussian i.i.d. negative control is rejected at 0%, so the test retains ample power to refuse clearly-wrong generators. The CHMM OoS KS pass rates of 79–86% sit just below the 90% ceiling (a genuine fidelity claim, not a low-power artefact); GARCH’s 58% is correctly rejected.

Closing the kurtosis gap with heavy-tailed emissions. CHMM-N produces excess kurtosis of 5.02 against an observed 7.68, a gap inherent to Gaussian emissions: a Gaussian mixture’s kurtosis is bounded by the variance of component means relative to their variances, and pushing tail states out degrades the central fit. CHMM-t addresses this directly through per-state Student-t emissions whose ν_k is updated by ECM one-dimensional search, allowing tail states to carry arbitrarily high individual kurtosis at the cost of an IS overshoot to 14.57. The unpenalised overshoot is a single-state artefact: a bracket sweep $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0\}$ shows the fit does not pile against the lower bracket (median $\nu_k = 50$, only 2/18 states near the lower edge), and an exponential $1/\nu_k$ shrinkage prior at rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 with a 1pp KS cost (full rate sweep in Appendix A.38). CHMM-L sits between CHMM-N and CHMM-t on the kurtosis axis without any shape parameter at all: per-component excess kurtosis of 3 (versus 0 Gaussian and ν -dependent Student-t) brings simulated IS kurtosis to 6.75, the closest match in the panel to observed 7.68. The three-way decision is concrete and is summarised in Table 5; all three variants’ 5–95% envelopes bracket observed VaR/ES on both windows.

Stationarity scope and the OoS cliff on NVDA and JPM. Cross-ticker generalization (Section 6.4) confirms that IS fidelity is preserved across diverse risk profiles, but at $K = 18$ OoS degradation concentrates on NVDA (55.8%) and JPM (49.8%): the IS-fixed transitions and emissions

Table 5. Variant decision guide. Recommended CHMM emission family by use-case priority. All three variants share the forward-backward scaffold and the quantile-based initialisation; the M-step is the only architectural difference.

Use-case priority	Recommended variant
Simplest closed-form fit; kurtosis fidelity secondary	CHMM-N
Cleanest IS kurtosis match; no shape parameter	CHMM-L
Per-state heavy tails for tail-conditional consumers	CHMM-t

cannot capture sector-specific regime shifts (the AI-driven NVDA regime; the rate-repricing cycle for financials). The pattern persists at every K in the sweep, so it is not a model-selection issue. Walk-forward quarterly refitting recovers ~ 15 percentage points of the JPM gap; the residual cliff is honest evidence that a richer extension (time-varying emission scales, or a Bayesian online transition-matrix update) is needed to fully close it.

The 2022 structural break. The five-window rolling-origin evaluation (Section 6.5) reproduces the headline pass rates on four of five windows; the 2022 window is an explicit counter-example with OoS KS collapsing to 0.2–0.4% because the rate-hike cycle is a structural break that the preceding eight years of near-zero rates do not contain. The headline is therefore single-window for SPY, with the rolling-origin panel as the most informative single piece of negative evidence on the stationarity scope.

Why the Student-t copula beats SIM on cross-asset fidelity. The Pipeline B Student-t copula on top of the CHMM-N marginals attains an off-diagonal correlation MAE of 0.027 on the six-asset universe, against 0.076 for a Single Index Model factor baseline on the same marginals (Section 6.6; full SIM and Gaussian copula panels in Appendix A). Two structural reasons. First, rank reordering preserves marginals by construction: the simulated path is a permutation of the stand-alone CHMM path, so its empirical CDF coincides with the CHMM marginal exactly, while SIM’s affine combination $\alpha_j + \beta_j G_{m,t} + \eta_{j,t}$ convolves the market factor with idiosyncratic residuals and distorts the marginal whenever $\beta_j \neq 1$ or residuals are heavy. Second, the Student-t copula’s $\nu^* = 6$ implies non-trivial joint tail dependence that injects rare concordant extreme moves the Gaussian copula systematically understates. The copula advantage is intrinsic to the rank-reordering construction rather than specific to the marginal family.

Filter-VaR over-conservatism is an open operational question. The filter-based regime-conditional VaR rule (Section 7.2) fails Kupiec on the 2024–2026 OoS window for all three CHMM emission families because the mixture of state-conditional emissions is systematically over-conservative; the inequality $F_{\sum_k \pi_k F_k}^{-1}(\alpha) \leq \min_k F_k^{-1}(\alpha)$ identifies the structural mechanism. A penalised-ECM $1/\nu_k$ shrinkage that brings simulated kurtosis from 14.30 to ≈ 8 leaves the filter-VaR Kupiec failure essentially unchanged, so the over-conservatism is a mixture-over-states property of the filter rather than a heavy-tails property of any one emission. We flag this as a limit on operational deployment, not as a contribution.

Limitations. The cross-ticker study covers six tickers; scaling to a larger universe via vine or factor copulas is a natural next step. State selection is via information-criterion plus distributional pass-rate rather than held-out log-likelihood; a clean re-selection on a non-OoS validation slice picks $K^* = 3$ by held-out log-lik (Appendix A), so the $K = 18$ operating point is best read as a multi-metric choice

that weights tail fidelity alongside likelihood. Skew-heavy-tailed emissions (skew- t or skew-Laplace), explicit-duration semi-Markov sojourns, and operational regime-conditional VaR are deferred as companion-paper directions.

9 Conclusion

A continuous hidden Markov model trained by Baum-Welch with quantile-based initialization reproduces the three Cont stylized facts of daily equity returns simultaneously at moderate state resolution; the closed-form mixture-of-eigenvalues identity for the absolute-return ACF recasts the Rydén et al. low- K failure as a K -rank statement on $\mathbf{T}$. Three emission variants share the same scaffold and land at three distinct points on the kurtosis axis: the Gaussian CHMM-N undershoots, the Student- t CHMM-t overshoots and is the per-state heavy-tailed choice for tail-conditional consumers, and the Laplace CHMM-L matches observed kurtosis cleanly without a shape parameter. The M-step swap is the only architectural difference across the three.

The headline quantitative findings on SPY are 93.8% in-sample and 84.2% out-of-sample Kolmogorov–Smirnov pass rates, an absolute-return ACF mean absolute error within 0.003 of GARCH(1,1), a clean Kupiec pass on Value-at-Risk at both 1% and 5%, and Value-at-Risk and Expected-Shortfall envelopes that bracket the observed historical values on both windows. These claims survive a rolling-origin generalisation on four of five windows; the 2022 rate-hike cycle is an explicit stationarity counter-example, and substantial OoS degradation on NVDA and JPM reflects the same scope limit on single names.

Multi-asset synthesis composes cleanly with the rank-based Student- t copula on top of the CHMM-N marginals: profile MLE selects $\nu^* = 6$, and the resulting cross-asset construction reproduces the off-diagonal correlation matrix on six SPY-member assets with a mean absolute error of 0.027, against 0.076 for a Single Index Model factor baseline on the same marginals.

Skew-heavy-tailed emissions, explicit-duration semi-Markov sojourns, and operational regime-conditional VaR are companion-paper directions; scaling the cross-asset construction to a larger universe via vine or factor copulas is the natural multi-asset follow-up.

Data and code availability. The simulation scripts and data to reproduce the paper are available under an MIT license at <https://github.com/altashly1/CHMM-paper>. The companion Julia package `CHMM-Model.jl` is released alongside this paper at <https://github.com/altashly1/CHMM-Model> and ships as a co-equal deliverable: the public API listed in Appendix A is the entry point that regenerates every table and figure from a single seed root under Julia ≥ 1.12 .

Conflict of interest. The authors declare no competing interests.

Author contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, conducted the empirical analysis, and generated all figures and tables. C.J. contributed to methodology review and validation. All authors edited and reviewed the final manuscript.

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A Supplementary Material

This appendix collects supplementary material referenced in the main body. Section [A.1](#) gives the public API and reproducer entry point of the companion Julia package `CHMM-Model.jl`. Section [A.2](#) gives the EM forward-backward recursions and the algorithm pseudocode. Section [A.6](#) gives the precise validation metric definitions. Section [A.10](#) states the formal propositions cited inline in Section [5](#) (ECM monotonicity, identifiability, MLE consistency, rank-reordering marginal preservation, filter-VaR mixture-quantile bound). Section [A.11](#) gives the full state-resolution and multi-emission sensitivity panels, the $K = 2$ replication, and the per- K figures. Section [A.24](#) gives KS test-power calibration, the block-bootstrap KS recalibration, multi-seed Monte Carlo, walk-forward refit, and the block-bootstrap baseline; price-fan figures and an expanded GARCH panel are included for

completeness. Section A.33 gives the full cross-asset panel including SIM and Gaussian copula, with copula derivations and profile-MLE figure. Section A.38 collects the pre-OoS validation K -selection, the walk-forward summary, and the ν_k -bracket / penalised-ECM diagnostics referenced from the discussion.

A.1 CHMM-Model.jl Public API and Reproducer

The companion Julia package `CHMM-Model.jl` (<https://github.com/altashly1/CHMM-Model>) is the entry point used to regenerate every table and figure in this paper from a single seed root under Julia ≥ 1.12 . The public surface is a single `CHMM` type, three emission constructors, a unified `fit!` method (log-space Baum-Welch / ECM), and a unified `simulate` method.

```
using CHMM
chmm_n = make_chmm_n(returns; K = 18)      # Gaussian emissions
chmm_t = make_chmm_t(returns; K = 18)      # Student-t emissions
chmm_l = make_chmm_l(returns; K = 18)      # Laplace emissions
fit!(chmm_n; max_iter = 60, tol = 1e-4)    # log-space Baum-Welch
paths = simulate(chmm_n; T = 2516, n = 1000) # 1000 paths of length T
```

The same `simulate` interface is used for every benchmark generator (Bootstrap, Gaussian i.i.d., Laplace i.i.d., GARCH(1,1)) so that the seven-metric panel evaluates each row on byte-identical windows. The Pipeline B Student-t copula sampler is exposed as `simulate_copula(...; nu = 6)` on the per-asset CHMM marginals; the rank-reordering construction is one function call. Running `julia -project=. run_full_rebuild.jl` from the paper repository (<https://github.com/altashly1/CHMM-paper>) reproduces every numerical artifact under the global seed policy of Section 3.5; `Manifest.toml` pins the dependency graph.

A.2 CHMM Algorithms and Derivations

A.3 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [9, 33]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1), \quad (9)$$

$$\log \alpha_t(j) = \log \sum_i \exp(\log \alpha_{t-1}(i) + \log T_{ij}) + \log b_j(O_t), \quad t = 2, \dots, T, \quad (10)$$

$$\log \beta_T(k) = 0, \quad (11)$$

$$\log \beta_t(i) = \log \sum_j \exp(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \dots, 1, \quad (12)$$

where $\log \text{sumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i, j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k) \beta_t(k)}{\sum_j \alpha_t(j) \beta_t(j)}, \quad \xi_t(i, j) = \frac{\alpha_t(i) T_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m) T_{mn} b_n(O_{t+1}) \beta_{t+1}(n)}, \quad (13)$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k) O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k) (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \quad (14)$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [10], Liu and Rubin [11] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \quad (15)$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \quad (16)$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q -function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee on the compact bracket $[\nu_{\min}, \nu_{\max}]$, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) = \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k) - \lambda / \nu, \quad (17)$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [10], Liu and Rubin [11] ECM; the rate sweep and its effect on simulated IS kurtosis are in Section 8).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $b_k(x) = \frac{1}{2b_k} \exp(-|x - \mu_k|/b_k)$ the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T}; \gamma_{1:T}(k)), \quad b_k^{\text{new}} = \frac{\sum_t \gamma_t(k) |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \quad (18)$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

A.4 Algorithm Descriptions

This section provides detailed pseudocode for the main algorithms used in this study. Algorithm 1 is the unified EM procedure shared across all three emission families. It makes explicit the *shared scaffold* (quantile-based initialization, log-space forward-backward, transition update) and the three *family-specific branches* (Gaussian M-step for CHMM-N, ECM with one-dimensional golden-section ν -search for CHMM-t, closed-form weighted-median + weighted-MAD for CHMM-L); the branch points are lines 3, 12, and 28. Visualizing the three variants as a single algorithm with three M-step branches makes the architectural contribution of the paper (one EM engine, three interchangeable emission families) immediate from the pseudocode. Algorithm 2 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 3 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 4 describes the cross-asset rank-reordering simulator for copula-based multi-asset synthesis. Algorithm 5 details the SIM regression-and-resampling pipeline.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 12, and 28 branch on the emission family $F \in \{\mathcal{N}, t, L\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t instantiates an ECM sequence whose second CM step updates ν_k by a one-dimensional golden-section search on the per-state Q -function.

Require: Observations $O_{1:T}$, number of states K , tolerance tol , max_iter , emission family $F \in \{\mathcal{N}, t, L\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$.

Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\theta_{1:K}$, initial distribution π .

```

1: Quantile-Based Initialization.
2: Sort  $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$  and partition into  $K$  equal chunks  $\{C_1, \dots, C_K\}$ .
3: for  $k = 1$  to  $K$  do
4:   switch  $F$ 
5:      $F = \mathcal{N}$ :  $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ .
6:      $F = t$ :  $\mu_k \leftarrow \text{mean}(C_k)$ ,  $\sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$ ,  $\nu_k \leftarrow \nu^{(0)}$ .
7:      $F = L$ :  $\mu_k \leftarrow \text{median}(C_k)$ ,  $b_k \leftarrow \max(\text{mean}(|C_k - \mu_k|), 10^{-6})$ .
8:   end for
9:  $T_{ij} \leftarrow 1/K$ ;  $\pi_k \leftarrow 1/K$ ;  $\mathcal{L}_{\text{prev}} \leftarrow -\infty$ .
10: EM Loop.
11: for  $n = 1$  to  $\text{max\_iter}$  do
12:   E-Step, per-family emission log-likelihood.
13:   switch  $F$ 
14:      $F = \mathcal{N}$ :  $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t | \mu_k, \sigma_k^2)$ .
15:      $F = t$ :  $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$ .
16:      $F = L$ :  $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$ .
17:   E-Step, shared log-space forward-backward.
18:    $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$ .
19:   for  $t = 2$  to  $T$ ,  $j = 1$  to  $K$  do
20:      $\log \alpha_t(j) \leftarrow \log \sum_i \exp(\log \alpha_{t-1}(i) + \log T_{ij} + \log B_{t,j})$ .
21:   end for
22:    $\log \beta_T(k) \leftarrow 0$ .
23:   for  $t = T - 1$  down to  $1$ ,  $i = 1$  to  $K$  do
24:      $\log \beta_t(i) \leftarrow \log \sum_j \exp(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$ .
25:   end for
26:   Compute  $\gamma_t(k)$  and  $\xi_t(i, j)$  from equation (13).
27:   Shared transition update.  $T_{ij} \leftarrow \sum_t \xi_t(i, j) / \sum_t \gamma_t(i)$ ;  $\pi_k \leftarrow \gamma_1(k)$ .
28:   M-Step, per-family emission update.
29:   switch  $F$ 
30:      $F = \mathcal{N}$  {classical Baum-Welch [8].}
31:      $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ .
32:      $\sigma_k \leftarrow \max(\sqrt{\sum_t \gamma_t(k) (O_t - \mu_k)^2 / \sum_t \gamma_t(k)}, 10^{-6})$ .
33:      $F = t$  {ECM of Peel and McLachlan [10], Liu and Rubin [11]; closed-form  $(\mu_k, \sigma_k)$  given  $\nu_k$ , then
        1D search on  $\nu_k$ .}
34:      $u_{t,k} \leftarrow (\nu_k + 1) / (\nu_k + ((O_t - \mu_k) / \sigma_k)^2)$  for all  $t, k$ .
35:      $\mu_k \leftarrow \sum_t \gamma_t(k) u_{t,k} O_t / \sum_t \gamma_t(k) u_{t,k}$ .
36:      $\sigma_k \leftarrow \max(\sqrt{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2 / \sum_t \gamma_t(k)}, 10^{-6})$ .
37:      $\nu_k \leftarrow \arg \max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_{\nu}(O_t; \mu_k, \sigma_k)$  {40-iter golden-section search.}
38:      $F = L$  {closed-form weighted Laplace MLE.}
39:      $\mu_k \leftarrow \text{WeightedMedian}(O_{1:T}, \gamma_{1:T}(k))$  {cumulative-weight bracketing with interpolation at the
         $W_k/2$  crossing.}
40:      $b_k \leftarrow \max(\sum_t \gamma_t(k) |O_t - \mu_k| / \sum_t \gamma_t(k), 10^{-6})$ .
41:   Convergence check.  $\mathcal{L}^{(n)} \leftarrow \log \sum_k \exp(\log \alpha_T(k))$ .
42:   if  $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$  then
43:     break
44:   end if
45:    $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$ .
46: end for
47: return  $\mathbf{T}, \theta_{1:K}, \pi$   $\{\theta_k = (\mu_k, \sigma_k)$  for  $F = \mathcal{N}$ ;  $(\mu_k, \sigma_k, \nu_k)$  for  $F = t$ ;  $(\mu_k, b_k)$  for  $F = L\}$ 

```

Algorithm 2 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$

Ensure: Most probable state sequence $S_{1:T}^*$

```
1: for  $k = 1$  to  $K$  do
2:    $\log \delta_1(k) \leftarrow \log(1/K) + \log \mathcal{N}(O_1 \mid \mu_k, \sigma_k^2)$ .
3: end for
4: for  $t = 2$  to  $T$  do
5:   for  $j = 1$  to  $K$  do
6:      $\text{vals}(j) \leftarrow \log \delta_{t-1}(i) + \log T_{ij}$  for all  $i$ .
7:      $\log \delta_t(j) \leftarrow \max_i \text{vals}(i) + \log \mathcal{N}(O_t \mid \mu_j, \sigma_j^2)$ .
8:      $\psi_t(j) \leftarrow \arg \max_i \text{vals}(i)$ .
9:   end for
10: end for
11:  $S_T^* \leftarrow \arg \max_k \log \delta_T(k)$ .
12: for  $t = T - 1$  down to  $1$  do
13:    $S_t^* \leftarrow \psi_{t+1}(S_{t+1}^*)$ .
14: end for
15: return  $S_{1:T}^*$ 
```

Algorithm 3 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M , Number of paths P

Ensure: Simulated return paths $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$

```
1: for  $p = 1$  to  $P$  do
2:   Sample initial state  $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ .
3:   for  $t = 1$  to  $M$  do
4:     Emit observation  $\hat{G}_t^{(p)} \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$ .
5:     if  $t < M$  then
6:       Transition to next state  $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t, \cdot})$ .
7:     end if
8:   end for
9: end for
10: return  $\{\hat{G}_{1:M}^{(p)}\}_{p=1}^P$ 
```

A.5 CHMM-t Degrees-of-Freedom Diagnostics

This appendix backs the discussion in Section 8 of the CHMM-t IS kurtosis overshoot. Figure 5 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit on SPY IS (seed = 20260420). Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

Table 6 reports the seven-metric panel for CHMM-t fits whose ECM bracket lower bound is lifted from $\nu_{\min} = 2.1$ through $\{2.5, 3.0, 4.0\}$, plus the Gaussian limit ($\nu = \infty$, equivalent to CHMM-N). Lifting the bracket does not reduce the IS kurtosis overshoot: the simulated kurtosis moves within the 12.5–14.0 band across bracket floors and only collapses to 5.05 in the full Gaussian limit. The overshoot is therefore a feature of the Student-t emission family at $K = 18$ and not a lower-bracket artefact.

Algorithm 4 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (PSD, unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T , number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

```
1: for  $p = 1$  to  $P$  do
2:   Draw copula sample  $\mathbf{U}^{(p)} \in [0, 1]^{T \times d}$  from  $\mathcal{C}(\Sigma)$ :
3:    $L \leftarrow \text{Cholesky}(\Sigma)$ ,  $Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$  i.i.d.
4:    $Y \leftarrow ZL^\top$ 
5:   if  $\mathcal{C} = \text{Gaussian}$  then
6:      $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$  componentwise
7:   else
8:     Draw  $W \sim \chi_\nu^2/\nu$  i.i.d. for  $t = 1, \dots, T$ ;  $X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$ 
9:      $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$  componentwise
10:  end if
11:  for  $j = 1$  to  $d$  do
12:    Simulate  $\tilde{g}_{j,1:T}^{(p)}$  from  $\mathcal{M}_j$  via Algorithm 3.
13:    Compute order statistics  $\tilde{g}_{j,(1)}^{(p)} \leq \dots \leq \tilde{g}_{j,(T)}^{(p)}$ .
14:    Compute ranks  $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$  for  $t = 1, \dots, T$ .
15:     $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$  for  $t = 1, \dots, T$ .
16:  end for
17: end for
18: return  $\{\hat{G}_{j,t}^{(p)}\}$ 
```

Algorithm 5 Single Index Model (SIM) construction and simulation

Require: IS return matrix $\mathbf{R} \in \mathbb{R}^{T \times d}$, market-column index m , market CHMM $\mathcal{M}_m$, path length T , number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

```
1: OLS fit. For  $j \neq m$ :
2:    $\hat{\beta}_j \leftarrow \text{Cov}(\mathbf{R}_{:,j}, \mathbf{R}_{:,m})/\text{Var}(\mathbf{R}_{:,m})$ 
3:    $\hat{\alpha}_j \leftarrow \overline{\mathbf{R}_{:,j}} - \hat{\beta}_j \overline{\mathbf{R}_{:,m}}$ 
4:    $\hat{\eta}_{j,t} \leftarrow R_{t,j} - \hat{\alpha}_j - \hat{\beta}_j R_{t,m}$ ,  $R_j^2 \leftarrow 1 - \sum_t \hat{\eta}_{j,t}^2 / \sum_t (R_{t,j} - \overline{\mathbf{R}_{:,j}})^2$ 
5: for  $p = 1$  to  $P$  do
6:   Simulate market path  $G_{m,1:T}^{(p)}$  from  $\mathcal{M}_m$  via Algorithm 3.
7:    $\hat{G}_{m,t}^{(p)} \leftarrow G_{m,t}^{(p)}$ .
8:   for  $j \neq m$  do
9:     Draw residual indices  $\{\iota_t\}_{t=1}^T \sim \text{Uniform}(\{1, \dots, T\})$ .
10:     $\hat{G}_{j,t}^{(p)} \leftarrow \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,\iota_t}$ .
11:  end for
12: end for
13: return  $\{\hat{G}_{j,t}^{(p)}\}$ 
```

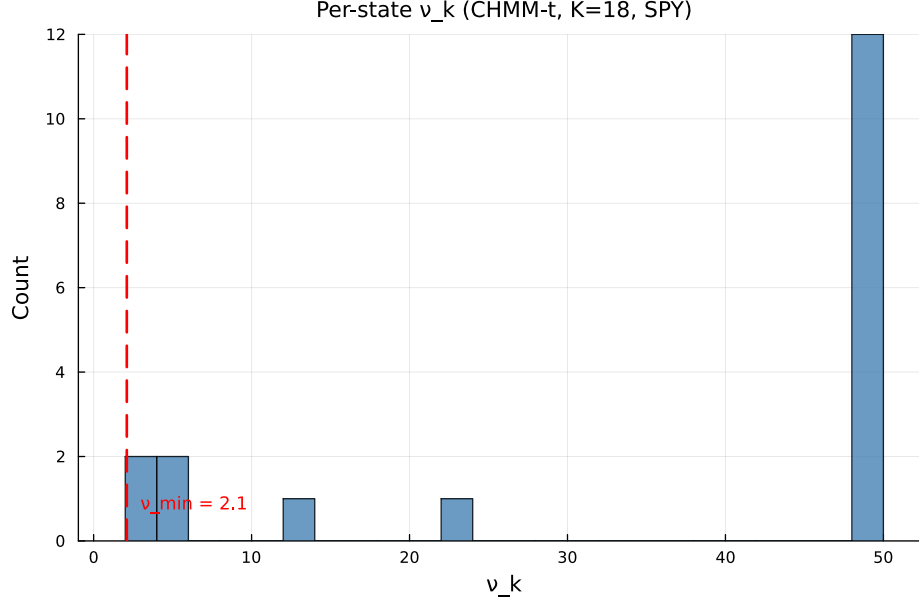


Figure 5. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

Table 6. CHMM-t $\nu_{\min}$ bracket sensitivity at $K = 18$ on SPY (seed = 20260420, 1,000 paths).

$\nu_{\min}$	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
2.1	95.7	90.2	86.2	82.0	13.95	10.08	0.0546	0.100	0.0759
2.5	95.8	90.6	84.2	79.1	15.12	10.34	0.0547	0.100	0.0759
3.0	93.8	89.5	85.3	80.4	12.25	9.65	0.0548	0.099	0.0760
4.0	95.5	89.5	85.0	80.3	13.25	9.39	0.0550	0.101	0.0759
∞ (Gauss)	94.1	86.7	84.2	75.5	5.03	4.46	0.0510	0.110	0.0809
<i>Observed</i>	–	–	–	–	7.68	5.29	–	–	–

A.6 Validation Metric Details

A.7 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [30, 31] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \quad (19)$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\rho}_{|G|}^{\text{sim}}(\tau) \right|. \quad (20)$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [41],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| du. \quad (21)$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^B (\sqrt{p_i} - \sqrt{q_i})^2}, \quad (22)$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}_{\text{sim}}^{(p)} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of Section 6.6), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P} \sum_{p=1}^P \frac{2}{d(d-1)} \sum_{1 \leq i < j \leq d} \left| \hat{\Sigma}_{\text{sim}, ij}^{(p)} - \hat{\Sigma}_{\text{obs}, ij} \right|.$$

A.8 Extended Evaluation: Detail Panels

Per-generator detail for the leverage, aggregational-Gaussianity, and joint p -value diagnostics on the SPY IS and OoS windows.

Table 7. Leverage effect (Pipeline A, SPY). Observed IS $\text{avg}_{k=1}^{20} \text{corr}(r_t^2, r_{t-k}) = -0.088$ with down-vs-up asymmetry 0.358; observed OoS averages are -0.066 and 0.478 . p_{avg} is the two-sided simulation-based coverage p -value for the observed average over 1,000 paths.

Model	avg IS	asym IS	p_{avg} IS	avg OoS	asym OoS	p_{avg} OoS
Bootstrap	0.000	0.001	0.000	0.001	-0.006	0.000
Gaussian	-0.000	-0.001	0.000	-0.000	0.001	0.000
Laplace	-0.000	-0.000	0.000	-0.000	0.005	0.000
GARCH(1,1)	-0.001	0.064	0.014	-0.001	0.065	0.061
CHMM-N	-0.038	0.292	0.000	-0.033	0.287	0.205
CHMM-t	-0.025	0.281	0.000	-0.029	0.274	0.048
CHMM-L	-0.024	0.322	0.000	-0.025	0.310	0.019

Table 8. Aggregational Gaussianity (Pipeline A, SPY). Median simulated kurtosis at aggregation horizons $h \in \{1, 5, 10, 21\}$ over 1,000 paths. Observed IS kurtosis is $\{7.68, 4.34, 6.82, 2.63\}$; observed OoS kurtosis is $\{5.29, 1.98, 0.09, -0.16\}$.

Model	IS kurtosis				OoS kurtosis			
	$h = 1$	$h = 5$	$h = 10$	$h = 21$	$h = 1$	$h = 5$	$h = 10$	$h = 21$
Observed	7.68	4.34	6.82	2.63	5.29	1.98	0.09	-0.16
Bootstrap	7.44	1.31	0.54	0.13	5.24	0.77	0.04	-0.39
Gaussian	-0.01	-0.03	-0.09	-0.17	-0.05	-0.17	-0.30	-0.54
Laplace	2.86	0.51	0.16	-0.04	2.55	0.23	-0.15	-0.49
GARCH(1,1)	4.35	4.79	3.74	2.55	1.79	1.87	1.26	0.34
CHMM-N	4.88	3.70	3.27	2.38	4.08	2.38	1.60	0.40
CHMM-t	8.46	2.65	2.06	1.35	5.55	1.89	1.15	0.21
CHMM-L	6.33	2.45	1.80	1.02	5.49	1.97	1.04	0.07

Table 9. Joint simulation-based p -value coverage (Pipeline A, SPY). Each cell is the two-sided simulation-based p -value for the observed statistic under the generator. $\bar{p}v$ is the mean across $\{\text{kurt}, \text{avg leverage}, \text{agg kurt } h = 5, \text{agg kurt } h = 21\}$; larger $\bar{p}v$ means better joint coverage.

Model	p_{kurt}	p_{acf}	p_{lev}	p_{ak5}	p_{ak21}	$\bar{p}v$ IS	$\bar{p}v$ OoS
Bootstrap	0.894	0.000	0.000	0.012	0.005	0.228	0.466
Gaussian	0.000	0.000	0.000	0.000	0.002	0.000	0.116
Laplace	0.002	0.000	0.000	0.000	0.002	0.001	0.156
GARCH(1,1)	0.271	0.021	0.014	0.903	0.974	0.540	0.468
CHMM-N	0.009	0.000	0.000	0.659	0.894	0.390	0.539
CHMM-t	0.872	0.000	0.000	0.196	0.228	0.324	0.661
CHMM-L	0.369	0.000	0.000	0.061	0.130	0.140	0.692

A.9 VaR / ES Envelope Visualization (Companion to Table 4)

Figure 6 visualizes the median and $[5\%, 95\%]$ envelopes across simulated paths whose numeric values are reported in Table 4.

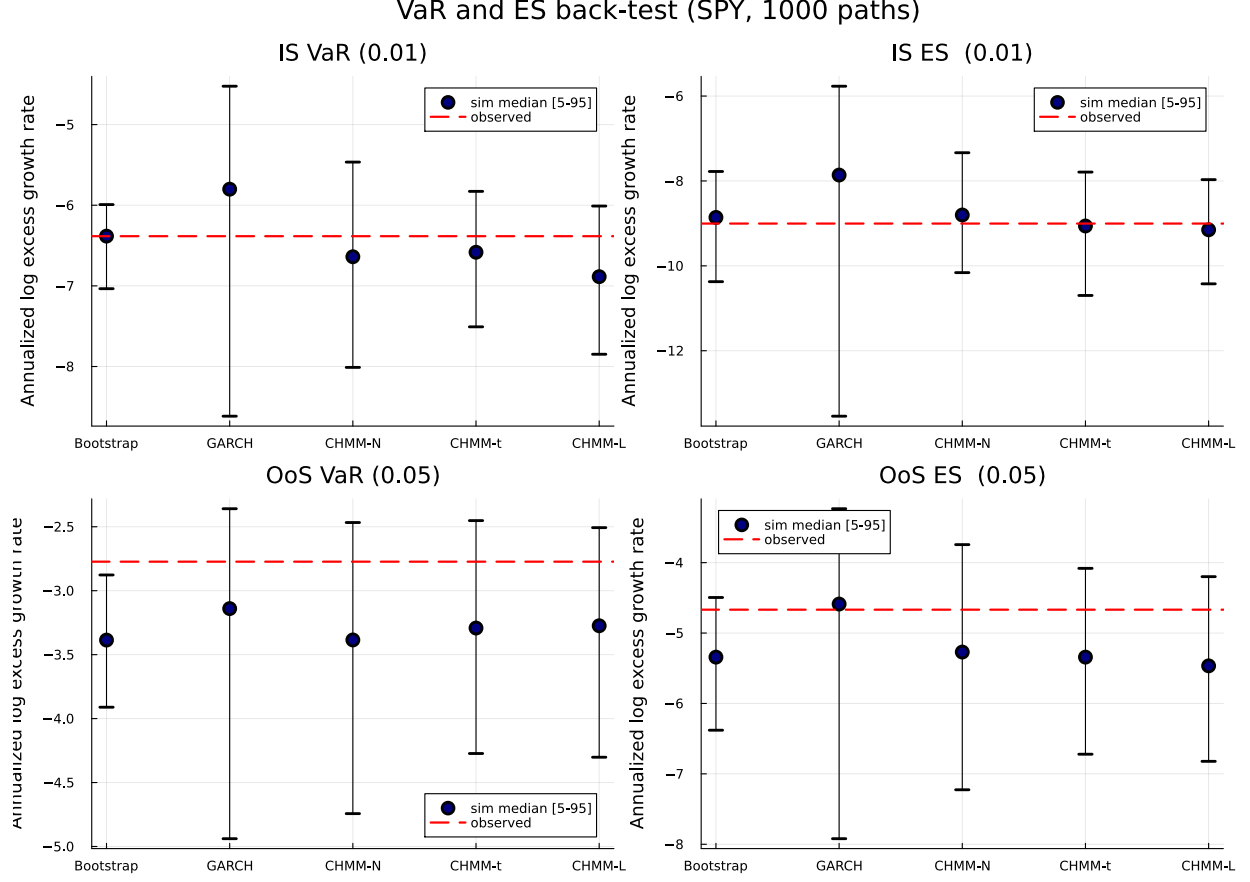


Figure 6. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1,000$; global seed policy in Section 3.5). Each subplot shows the simulated median (dot) and [5%, 95%] envelope (error bar) across paths against the observed historical value (dashed red line); rows are IS and OoS windows, columns are VaR and ES, and the x-axis lists the five generators (Bootstrap, GARCH(1,1), CHMM-N, CHMM-t, CHMM-L). The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Headline: every observed red line falls inside the corresponding simulated envelope, confirming that the CHMM family passes the coverage leg of the back-test complementary to the Kupiec / Christoffersen likelihood-ratio leg in Section 7.1.

A.10 Formal Theoretical Statements

The following propositions are referenced in Section 5 and stated formally here for reference; each is a one-line specialisation of a standard result and the proof sketch is one sentence.

Proposition 1 (ECM monotonicity for CHMM-t (and EM monotonicity for CHMM-N, CHMM-L)). *Under compactness of the parameter space (μ_k, σ_k or b_k, ν_k each in a closed bounded interval, $T_{ij} \geq \epsilon_T > 0$), the two-block CM step for CHMM-t of Section 4.2, namely the closed-form weighted updates of $(\mathbf{T}, \boldsymbol{\pi}, \mu_k, \sigma_k)$ at fixed ν_k , followed by a golden-section maximisation of the per-state Q -function over $\nu_k \in [\nu_{\min}, \nu_{\max}]$, satisfies $Q(\theta^{(n+1)} | \theta^{(n)}) \geq Q(\theta^{(n)} | \theta^{(n)})$, and consequently $\mathcal{L}(\theta^{(n+1)}) \leq \mathcal{L}(\theta^{(n)})$ where $\mathcal{L}$ is the incomplete-data log-likelihood. The same monotonicity holds for the closed-form CHMM-N and CHMM-L M-steps under the standard EM guarantee.*

Proof sketch. Each CM step does not decrease Q by definition (CM-1 is closed-form maximum, CM-2 is golden-section maximum); the standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \geq Q(\theta^{(n+1)} | \theta^{(n)})$

$\theta^{(n)} - Q(\theta^{(n)} | \theta^{(n)})$ [35, 36] converts Q -monotonicity into log-likelihood monotonicity. Compactness gives boundedness above, hence convergence.

Proposition 2 (Identifiability up to label permutation). *Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1) and the emission contrast condition (Assumption 2), the CHMM parameter is identifiable up to label permutation in all three emission families considered (CHMM-N, CHMM-t with $\nu_k \in [\nu_{\min}, \nu_{\max}]$, and CHMM-L).*

Proof sketch. The HMM identifiability theorem of Allman et al. [37] requires $\mathbf{T}$ of full rank and per-state emissions linearly independent in the space of densities on $\mathbb{R}$. Full-rank $\mathbf{T}$ follows from irreducibility [34]; linear independence of Gaussian, bracketed- ν Student-t, and Laplace emissions with distinct location-scale parameters follows from Yakowitz and Spragins [38].

Proposition 3 (MLE consistency). *Under Assumptions 1–2 and the assumption that the true parameter lies in the interior of a compact Θ , the MLE $\hat{\theta}_T = \arg \max_{\theta \in \Theta} \mathcal{L}_T(\theta)$ is consistent: $\hat{\theta}_T \xrightarrow{P} \theta_0$ as $T \rightarrow \infty$, modulo label permutation.*

Proof reference. Bickel et al. [39] (Theorem 1) and Douc et al. [40] apply directly under our assumption set: irreducibility supplies the mixing condition, finite second moment the moment condition, compactness the parameter-space condition.

Proposition 4 (Marginal preservation under rank reordering). *Let $\tilde{g}_{j,1:T}$ be a sample of size T from the per-asset CHMM marginal of asset j , and let the rank-reordered output be $\hat{g}_{j,t}^{(p)} = \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$ where $r_{j,t}^{(p)}$ is the rank of the copula sample. Then for every asset j the empirical CDF of $\{\hat{g}_{j,t}^{(p)}\}_{t=1}^T$ equals that of $\{\tilde{g}_{j,t}^{(p)}\}_{t=1}^T$.*

Proof sketch. The map $t \mapsto r_{j,t}^{(p)}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

Proposition 5 (Mixture-quantile lower bound, filter VaR). *Let $\pi \in [0, 1]^K$ be a probability vector, $F_1, \dots, F_K$ per-state CDFs, $F = \sum_k \pi_k F_k$, and $k^* = \arg \min_k F_k^{-1}(\alpha)$ with $q^* = F_{k^*}^{-1}(\alpha)$. Then $F(q^*) \leq \alpha$, with strict inequality whenever there exists $k \neq k^*$ with $\pi_k > 0$ and $F_k(q^*) < \alpha$. Consequently $F^{-1}(\alpha) \leq \min_k F_k^{-1}(\alpha)$ under the same condition.*

Proof sketch. At q^* , $F_{k^*}(q^*) = \alpha$ and $F_k(q^*) \leq \alpha$ for all k by monotonicity, hence $F(q^*) = \sum_k \pi_k F_k(q^*) \leq \alpha$, strict whenever some $F_k(q^*) < \alpha$ with $\pi_k > 0$. Monotonicity of F lifts the CDF inequality to the quantile.

A.11 State-Resolution and Multi-Emission Sensitivity

A.12 Empirical Stylized-Fact Visualization

Figure 7 visualizes the stylized facts of SPY daily excess growth rates summarised in Section 6.1 of the main text.

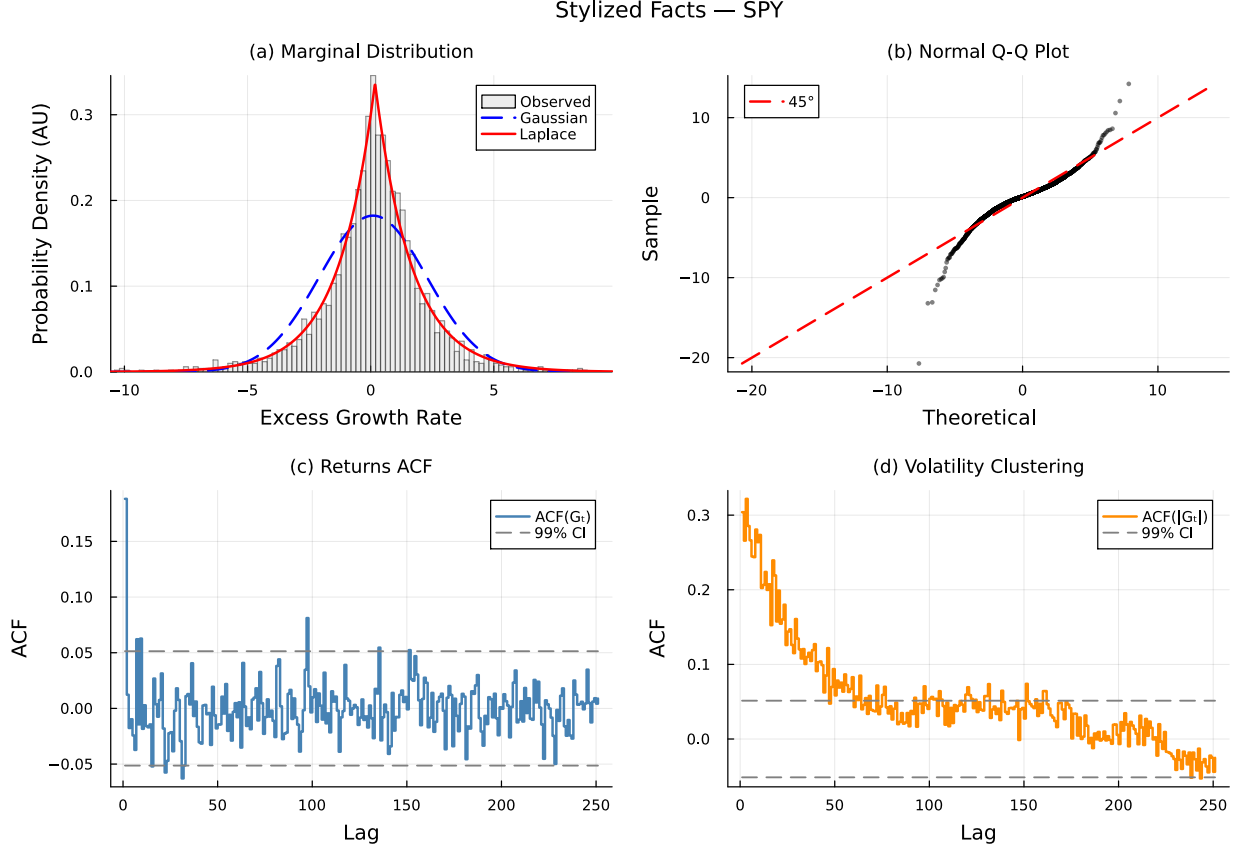


Figure 7. Empirical stylized facts of SPY daily excess growth rates (IS, 2014-01-03 through 2024-01-03). Panel (a): leptokurtic marginal distribution with Gaussian and Laplace fits. Panel (b): normal Q-Q plot confirming heavy tails. Panel (c): near-zero autocorrelation of raw returns. Panel (d): persistent autocorrelation of absolute returns characterizing volatility clustering.

A.13 Full State-Resolution Sensitivity Table

Table 10 reports the full K -sweep referenced in Section 6.2 of the main text. Observed excess kurtosis is 7.68 IS and 5.29 OoS at all K .

Table 10. State resolution sensitivity for SPY CHMM-N (1,000 paths, $\alpha = 0.05$). Every metric is computed on the identical simulation pipeline used for Table 1 in the main text. IS coverage was 100% at every K ; OoS coverage is reported separately.

K	KS (%)		AD (%)		Kurtosis		ACF-MAE	W_1	H	Cov OoS (%)	
	IS	OoS	IS	OoS	Obs	Sim					
3	88.8	77.1	79.2	67.9	7.68	3.80	0.0466	0.130	0.0836	94.9	
6	93.1	80.0	87.1	74.6	7.68	5.26	0.0499	0.113	0.0828	90.9	
9	94.6	82.4	85.8	73.6	7.68	4.56	0.0494	0.116	0.0806	93.9	
12	94.6	82.8	86.0	74.9	7.68	4.57	0.0495	0.112	0.0802	88.9	
15	93.9	82.7	86.9	75.6	7.68	4.89	0.0505	0.116	0.0806	90.9	
18	95.6	81.7	88.4	73.2	7.68	4.98	0.0509	0.110	0.0805	90.9	
21	96.1	86.6	89.0	78.4	7.68	3.83	0.0532	0.111	0.0798	100.0	

A.14 Multi-Emission State-Resolution Sensitivity

Table 11 extends Table 10 across the three emission families, confirming that the qualitative operating point $K = 18$ is stable when the Gaussian component is replaced with Student-t or Laplace emissions. For CHMM-t the per-state degrees of freedom ν_k are refit by ECM-plus-golden-section at every K ; for CHMM-L the weighted-median location and weighted MAD scale are closed form and impose no additional cost. Three patterns extend from Table 10 to the new families. First, the distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band for all three families: CHMM-t reaches its highest IS KS at $K = 18$ (95.2%) and its highest OoS KS at $K = 18$ (85.1%), and CHMM-L reaches its highest IS KS at $K = 18$ (96.2%) and its highest IS AD at $K = 18$ (93.0%). Once a family enters the plateau the operating point is flat with respect to K , so we report the main-text results at the same $K = 18$ used for CHMM-N. Second, the ACF-MAE remains within the narrow 0.047–0.056 band observed for the Gaussian sweep, indicating that the volatility-clustering mechanism is a property of the regime structure rather than the emission family. Third, simulated kurtosis behaves as expected from the component families: the Gaussian sweep peaks around 5.0 at the high- K end, the Laplace sweep lands squarely in the 5.2–6.6 range (closest to the observed 7.68), and the Student-t sweep climbs sharply above 13 across the entire sweep because the ECM search assigns small ν_k to a subset of tail states, over-correcting IS kurtosis.

Table 11. Multi-emission state resolution sensitivity for SPY (1,000 paths, $\alpha = 0.05$). CHMM-N: Gaussian; CHMM-t: Student-t with per-state ν_k refit by ECM-plus-golden-section; CHMM-L: Laplace with weighted-median location and weighted-MAD scale. Every metric is computed on the identical simulation pipeline used for Table 1. Observed excess kurtosis is 7.68 IS and 5.29 OoS.

Family	K	KS (%)		AD (%)		Kurt (IS sim)	ACF-MAE	W_1 IS	H IS
		IS	OoS	IS	OoS				
CHMM-N	3	89.8	76.3	81.3	67.7	3.81	0.0468	0.128	0.0835
	6	91.7	79.6	85.9	73.6	5.30	0.0497	0.115	0.0831
	9	92.8	82.2	84.7	75.5	4.55	0.0497	0.118	0.0810
	12	95.0	83.7	86.1	76.6	4.55	0.0497	0.114	0.0805
	15	95.2	83.5	86.9	76.3	4.91	0.0508	0.111	0.0802
	18	94.7	82.4	85.8	76.5	5.02	0.0509	0.111	0.0809
	21	95.2	87.6	89.1	80.9	3.80	0.0530	0.112	0.0802
CHMM-t	3	91.1	81.6	85.2	77.1	27.12	0.0540	0.109	0.0764
	6	92.2	82.1	86.1	76.6	20.80	0.0533	0.108	0.0768
	9	92.7	84.8	86.9	80.1	15.96	0.0537	0.104	0.0756
	12	94.1	83.1	88.3	80.6	14.68	0.0542	0.102	0.0757
	15	93.5	82.9	88.0	76.5	15.85	0.0537	0.103	0.0757
	18	95.2	85.1	90.9	80.7	14.40	0.0550	0.101	0.0756
	21	94.9	82.2	89.3	77.8	13.28	0.0540	0.103	0.0758
CHMM-L	3	82.4	63.6	82.9	66.9	5.24	0.0533	0.112	0.0853
	6	88.8	77.0	83.3	72.5	5.74	0.0511	0.113	0.0823
	9	90.0	77.9	84.1	74.8	6.05	0.0519	0.113	0.0823
	12	92.4	77.8	87.6	77.9	6.21	0.0540	0.106	0.0797
	15	92.6	79.4	87.0	77.6	6.06	0.0542	0.106	0.0795
	18	96.2	80.9	93.0	80.6	6.60	0.0564	0.101	0.0793
	21	90.7	80.7	87.3	78.8	5.99	0.0544	0.107	0.0809

A.15 Baum-Welch Convergence

Figure 8 shows the Baum-Welch convergence curves for representative state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.

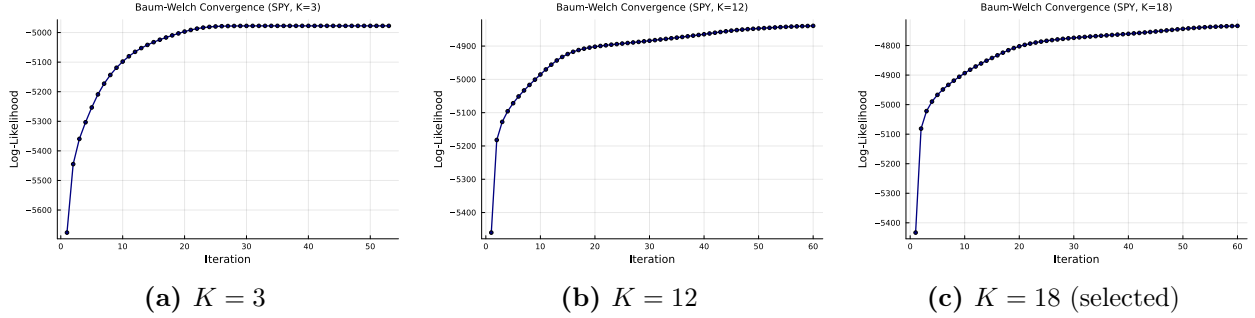


Figure 8. Baum-Welch convergence for the SPY CHMM-N fit (Pipeline A, IS) at selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` = 60.

A.16 Emission Distributions Across State Resolutions

Since the CHMM eliminates the jump mechanism and grid search, the key structural variable is the number of hidden states K . Figure 9 compares the learned emission distributions for $K \in \{3, 6, 12, 18\}$, illustrating how increasing state resolution refines the decomposition of the observed return distribution.

At $K = 3$, the model partitions the return space into three broad regimes: a crash state (large negative μ , high σ), a calm state (near-zero μ , low σ), and a boom state (positive μ , moderate σ). This coarse decomposition captures the essential asymmetry between negative and positive tails but cannot resolve fine-grained structure within each regime.

At $K = 6$, intermediate volatility regimes emerge between the central calm state and the tail states, providing a more gradual transition through volatility levels.

At $K = 12$ –18, the emission distributions form a dense coverage of the return space, with each state capturing a narrow band of returns. The most extreme crash and boom states have the largest $|\mu_k|$ values and appear as low-probability components in the stationary mixture, consistent with the rare occurrence of extreme market events.

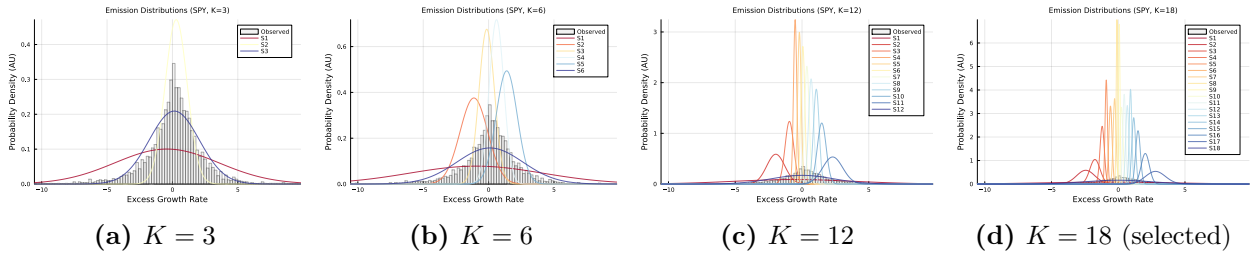


Figure 9. Learned CHMM-N (Gaussian) emission distributions at different state resolutions, overlaid on the observed SPY IS return histogram (Pipeline A). Higher K values produce finer decomposition with dedicated states capturing tail behavior.

A.17 Transition Matrix Structure

Figure 10 shows the learned transition matrices for $K = 6$ and $K = 18$ in $\log_{10}$ color scale. Both matrices exhibit a dominant near-diagonal band, reflecting the tendency of the market to remain in its current regime. Off-diagonal entries decay with distance from the diagonal, indicating that large regime jumps (for example, from a deep crash state directly to a boom state) are rare but nonzero.

The diagonal dominance is critical for the ACF-MAE result discussed in Section 8: the self-transition probabilities T_{kk} for central states are typically 0.7–0.95, producing mean sojourn times of 3–20 days. The mixture of these geometric sojourn times across states produces the sum-of-exponentials ACF profile that approximates the observed slow decay.

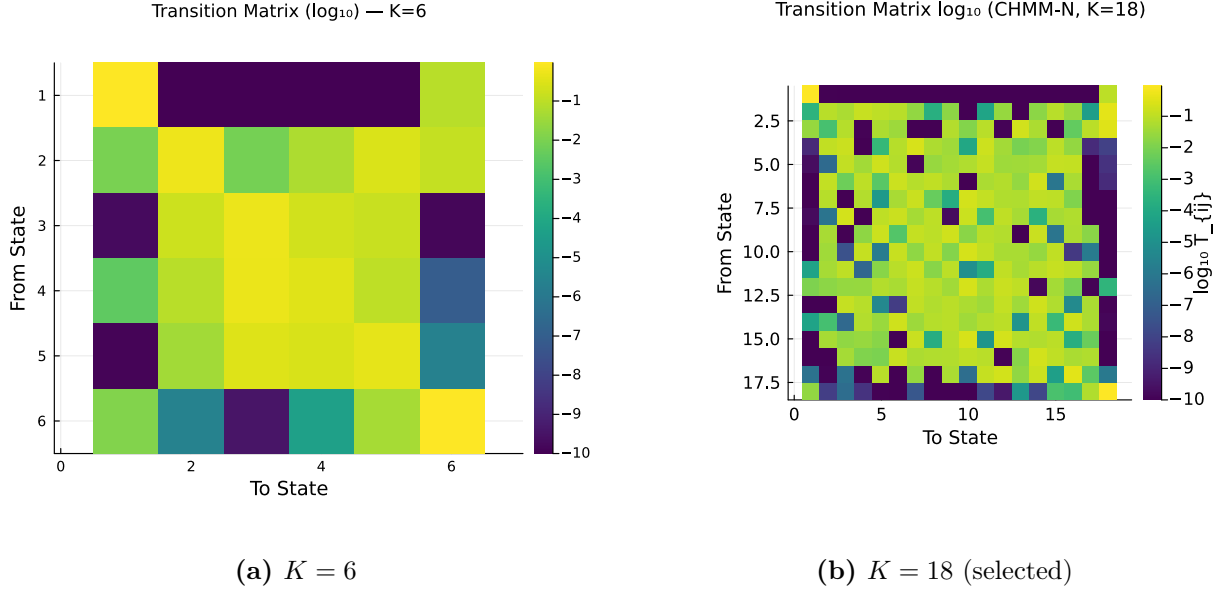


Figure 10. Learned CHMM-N transition matrices for the SPY IS fit (Pipeline A) at $K = 6$ and $K = 18$ in $\log_{10}$ color scale. The near-diagonal band reflects regime persistence; off-diagonal entries decay with distance, indicating that large regime transitions are rare.

A.18 Stationary Distributions and Residence Times

Figure 11 shows the stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 18$. In both cases, the central states (those with μ_k near zero and moderate σ_k) receive the highest stationary probability, consistent with the empirical concentration of returns near zero. Tail states (crash and boom) receive low stationary probability, consistent with the rarity of extreme events.

Figure 12 shows the natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for $K = 6$ and $K = 18$. Under pure Markov dynamics (no jump mechanism), the expected duration in state k before transitioning is τ_k steps. Central states exhibit the longest residence times (5–20 days), while tail states have shorter residence times (1–3 days), consistent with the transient nature of extreme market events. Notably, the tail states at higher K maintain sufficient self-transition probability ($T_{kk} \approx 0.3$ – 0.5) to produce multi-day tail episodes, the empirical correlate of the spectral mechanism in Section 5.

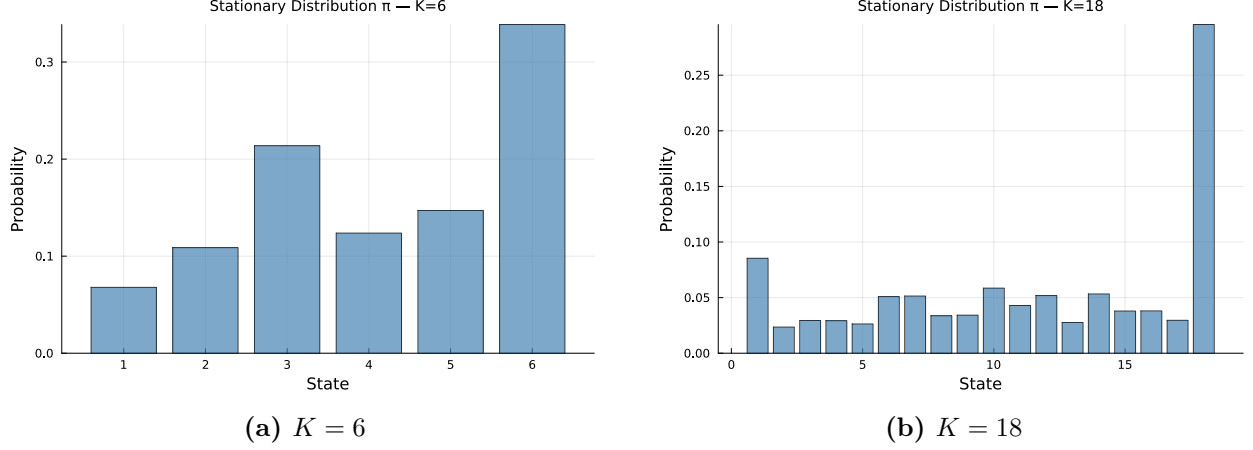


Figure 11. Stationary distributions $\bar{\pi}$ for the SPY CHMM-N IS fit (Pipeline A) at $K = 6$ and $K = 18$. Central states receive the highest stationary probability, consistent with the concentration of returns near zero.

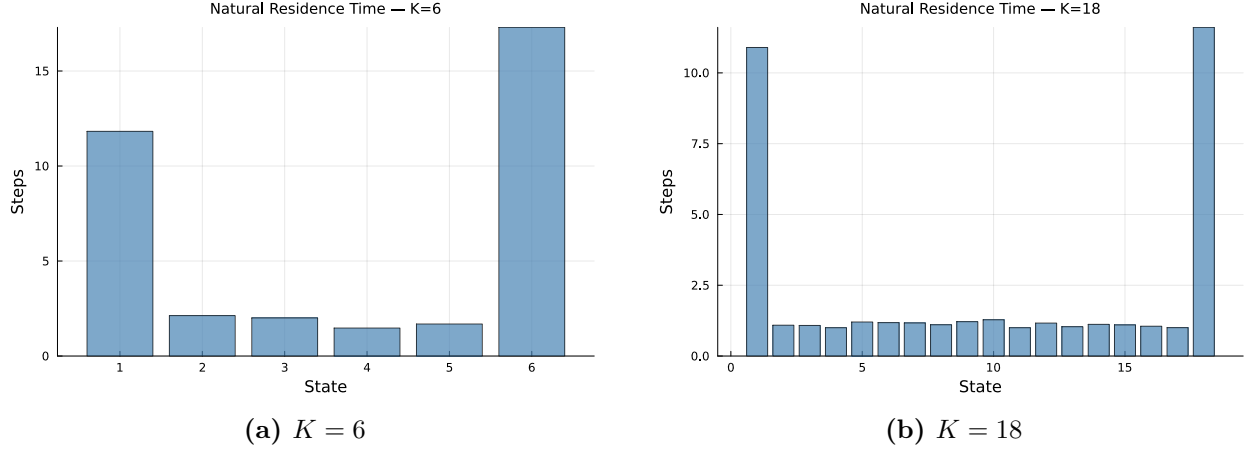


Figure 12. Natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for the SPY CHMM-N IS fit (Pipeline A) at $K = 6$ and $K = 18$. Central states have the longest residence times (5–20 days), while tail states are shorter-lived (1–3 days). This asymmetry drives volatility clustering.

A.19 IS Comparison at Non-Selected State Resolutions

Figures 13–16 show the IS model comparison panels for $K \in \{3, 6, 12, 21\}$, complementing the $K = 18$ comparison in Figure 3 of the main text. Across all resolutions, the CHMM’s simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern. The visible improvement from $K = 3$ to $K = 18$ is concentrated in the tails of the Q-Q panel and the steady-state level of the ACF panel. $K = 9$ and $K = 15$ panels are omitted because they interpolate smoothly between the shown resolutions; their numerical metrics are in Table 10.

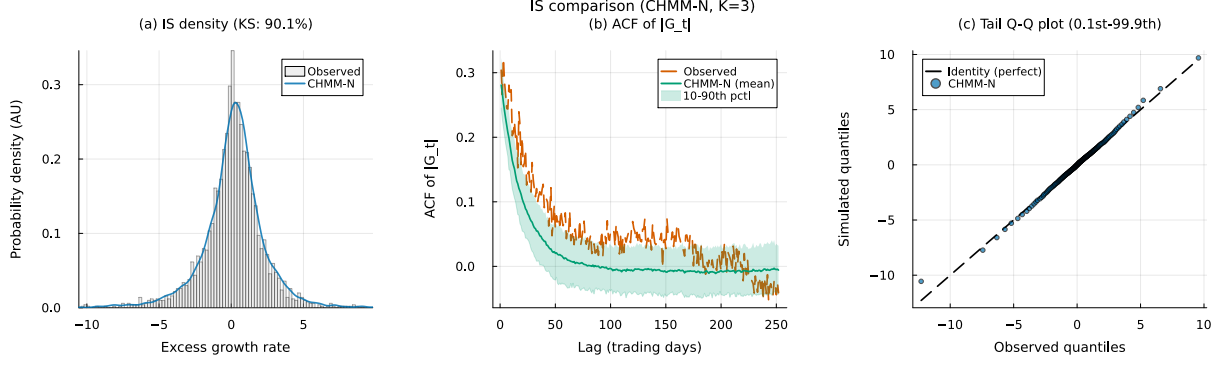


Figure 13. IS SPY CHMM-N comparison at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 3). The $K = 18$ counterpart is Figure 3 of the main text.

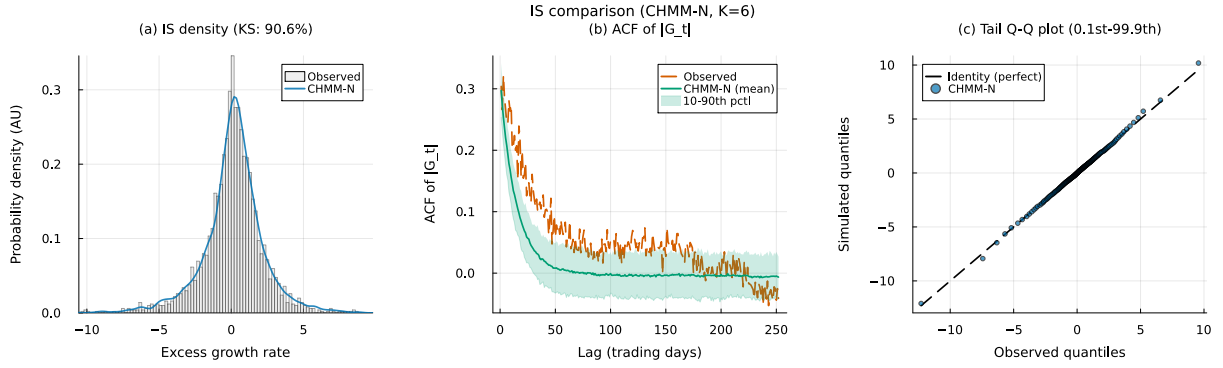


Figure 14. IS SPY CHMM-N comparison at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 3).

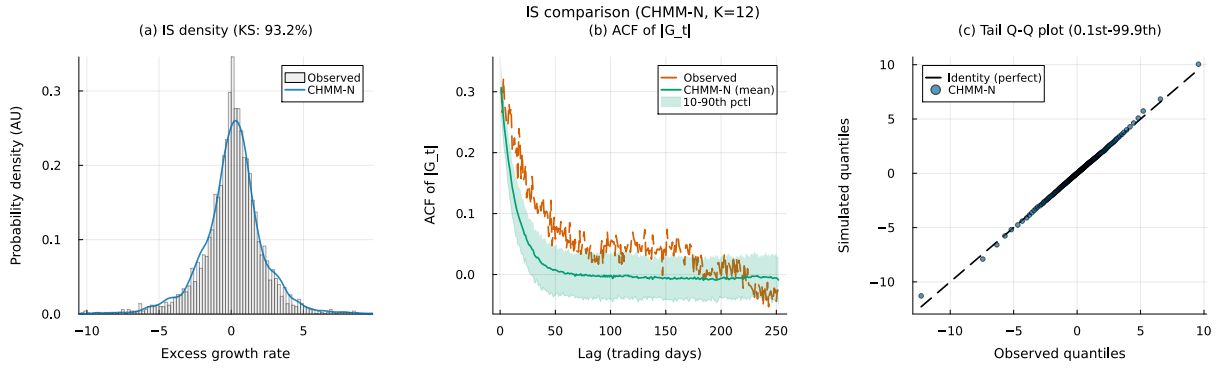


Figure 15. IS SPY CHMM-N comparison at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 3).

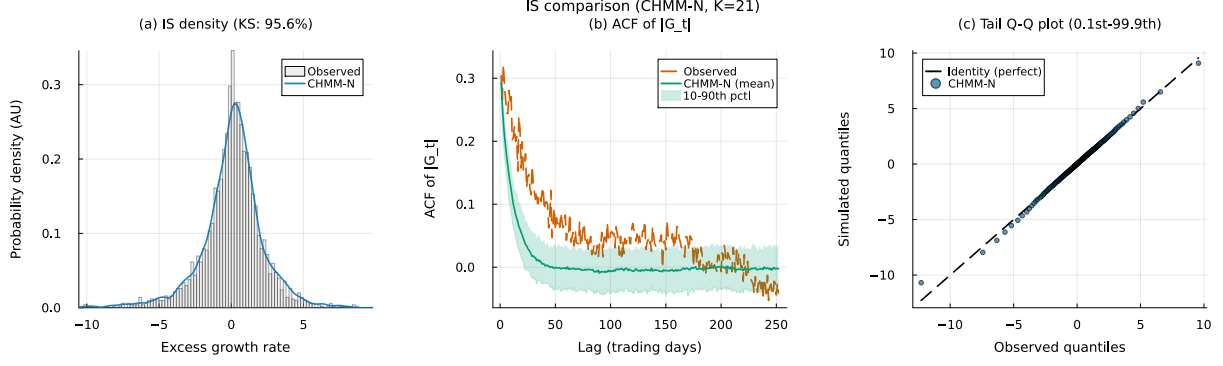


Figure 16. IS SPY CHMM-N comparison at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 3).

A.20 OoS Validation at Non-Selected State Resolutions

Figures 17–20 show the OoS validation panels for $K \in \{3, 6, 12, 21\}$. KS pass rates remain above 91% at all resolutions; the $K = 18$ panel is reproduced in Figure 4 of the main text. As with the IS panels, $K = 9$ and $K = 15$ are omitted; numerical OoS KS pass rates are in Table 10.

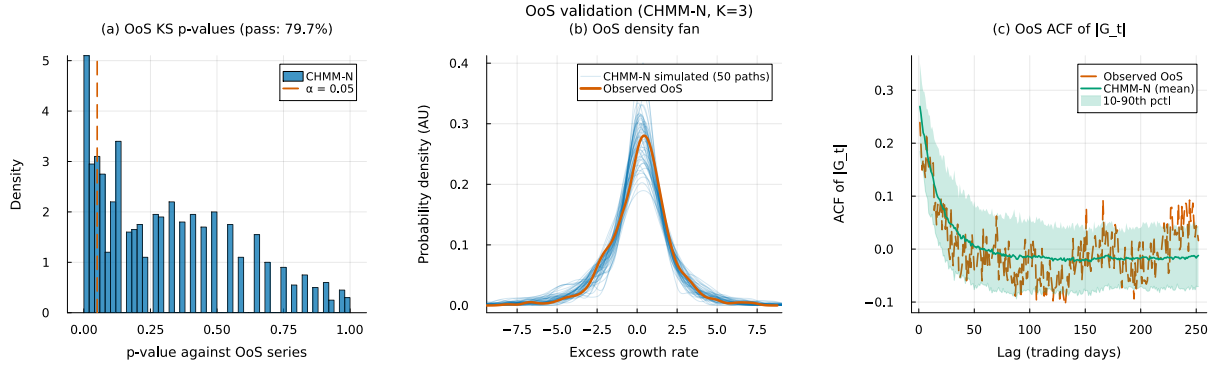


Figure 17. OoS SPY CHMM-N validation at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4). The $K = 18$ counterpart is Figure 4 of the main text.

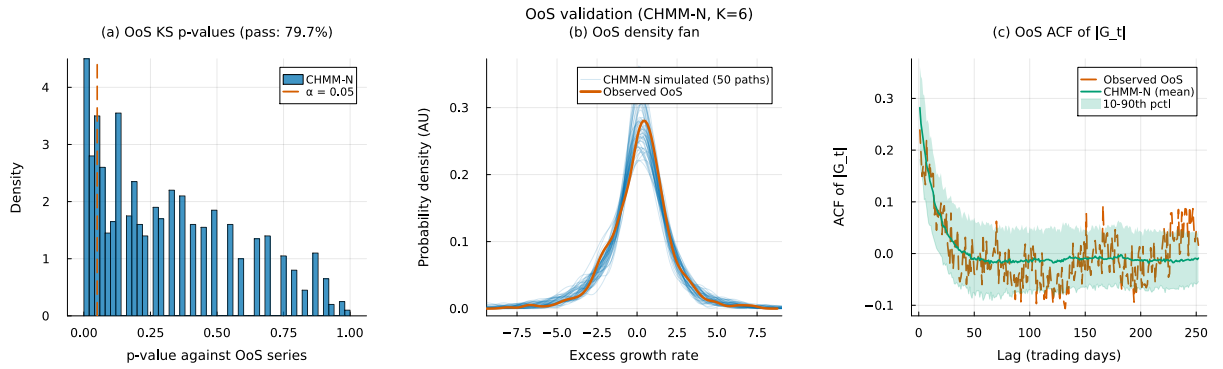


Figure 18. OoS SPY CHMM-N validation at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

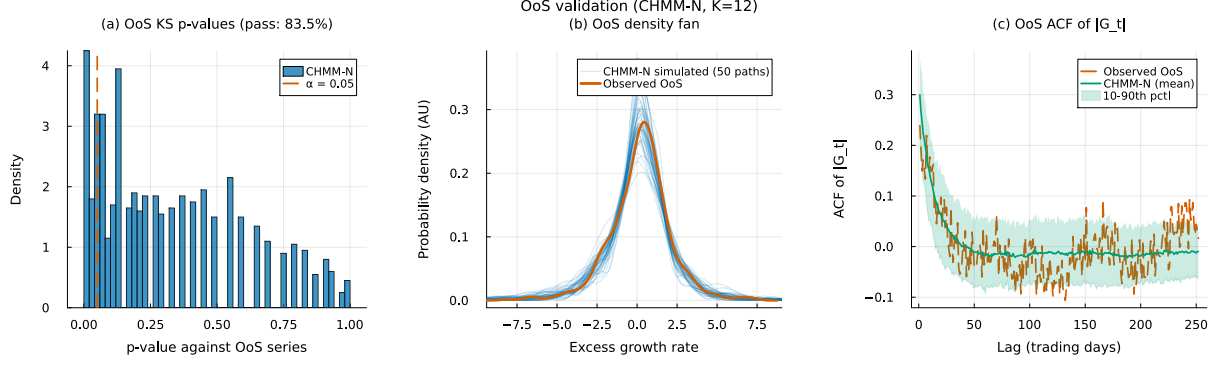


Figure 19. OoS SPY CHMM-N validation at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

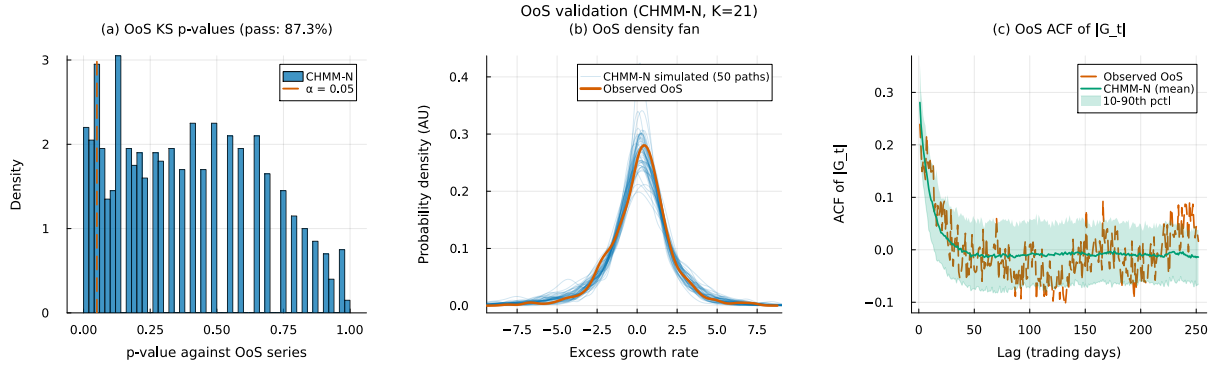


Figure 20. OoS SPY CHMM-N validation at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

A.21 Example Simulated Trajectories (Illustrative, IS-Fitted CHMM-N)

Figure 21 shows example simulated return trajectories from the SPY CHMM-N at $K = 6$ and $K = 18$, illustrating the qualitative dynamics produced by the model. Each panel overlays one unconditional simulated path (blue; the sampled-path index is labelled in the figure legend) against the first 500 trading days of the observed IS SPY excess growth-rate series (red). The two lines do not need to agree pointwise (simulated and observed hidden-state sequences are drawn independently); the intended read is qualitative, showing that both trajectories exhibit the characteristic pattern of financial returns: extended periods of low volatility punctuated by clusters of large-magnitude events.

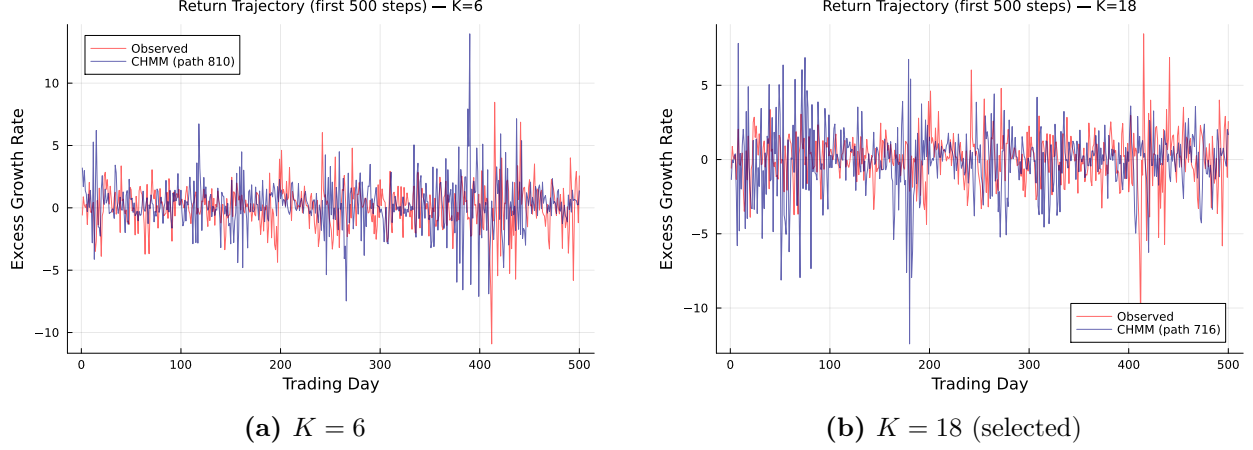


Figure 21. Example simulated return trajectories from the SPY CHMM-N (Pipeline A, IS fit, first 500 simulated steps). Blue: one unconditional simulated path (index labelled in the figure legend); red: first 500 trading days of the observed IS SPY excess growth rate series. The alternation between calm (central states) and volatile (tail states) periods produces the volatility clustering characteristic of empirical financial returns. This figure is qualitative: simulated and observed state sequences are independent, so point-by-point agreement is not expected.

A.22 Multi-Emission Figures at $K = 18$

This appendix reproduces the standard diagnostic suite (emission PDFs, residence times, IS comparison, OoS validation, and convergence) for the two heavy-tailed emission families (CHMM-t and CHMM-L) at the selected state resolution $K = 18$; transition matrices and stationary distributions are discussed in prose only below, since they are visually indistinguishable across families at $K = 18$. The Gaussian counterparts are already shown as Figures 3 and 4 in the main text and as Figures 8, 10, 11, and 12 above in this appendix. The figures confirm that the shared forward-backward scaffold produces internally-similar regime structure across the three families: the near-diagonal transition band, the K -component dense emission cover, the longer central-state residence times, and the monotone log-likelihood convergence are features of the CHMM framework and not of the Gaussian component specifically.

Emission PDFs. Figure 22 compares the learned $K = 18$ emissions for the three families. The CHMM-t panel’s per-state ν_k allocation is visible in the narrower modes of certain tail states; the CHMM-L panel’s Laplace components have sharper peaks and slower exponential tail decay than the Gaussians.

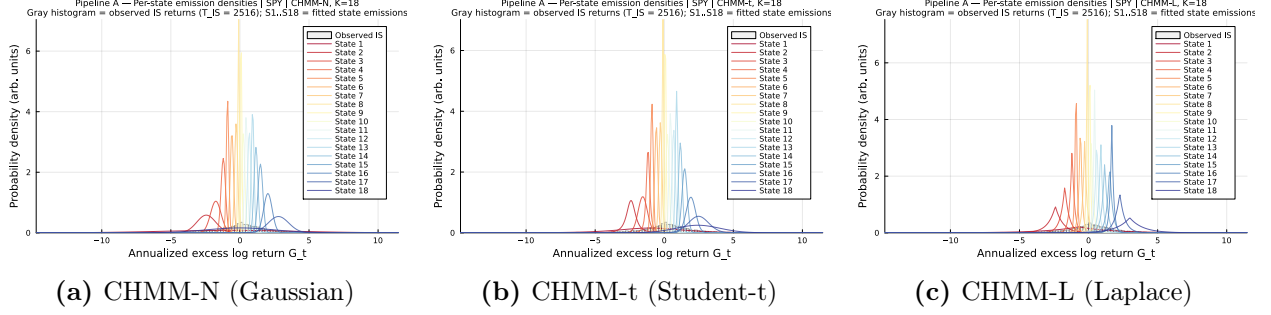


Figure 22. Learned emission distributions at $K = 18$ across the three emission families (CHMM-N Gaussian, CHMM-t Student-t with per-state ν_k , CHMM-L Laplace), overlaid on the observed SPY IS return histogram (Pipeline A). Heavier-tailed emission families (t, L) give individual components more tail mass, which is what closes the kurtosis gap of the Gaussian CHMM.

Transition matrices and stationary distributions. The $K = 18$ transition matrices and stationary distributions learned under CHMM-t and CHMM-L are visually indistinguishable from their CHMM-N counterparts (right-hand $K = 18$ panels of Figures 10 and 11): the near-diagonal band, the magnitude of self-transition probabilities, and the mass concentration at central states are features of the regime structure, not the emission family. This is consistent with the ACF-MAE being stable across the family axis (Table 11). The per-family residence times $\tau_k = 1/(1 - T_{kk})$, which magnify small differences in the self-transition probabilities each family recovers, are compared in Figure 23.

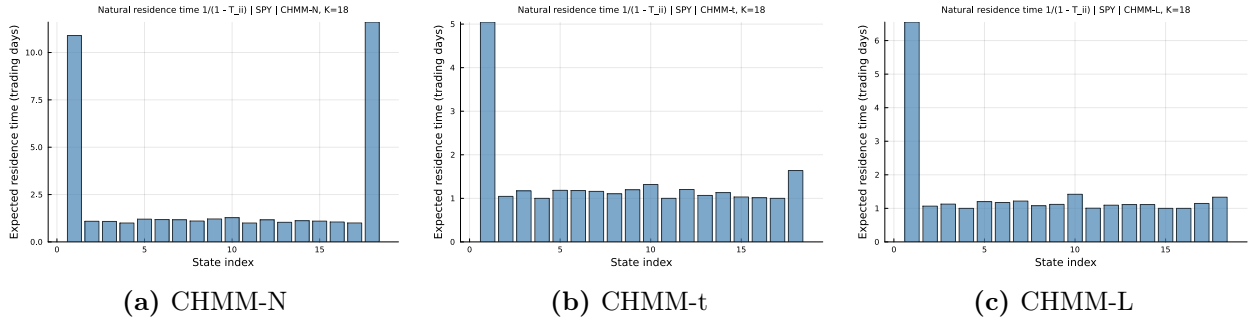


Figure 23. Natural residence times $\tau_k = 1/(1 - T_{kk})$ at $K = 18$ for the SPY IS fit (Pipeline A) across the three emission families (CHMM-N Gaussian, CHMM-t Student-t, CHMM-L Laplace).

IS comparison. Figures 24–25 reproduce the three-panel IS comparison (density, ACF of $|G_t|$, tail Q-Q) at $K = 18$ for the two heavy-tailed families; the CHMM-N panel is Figure 3 in the main text.

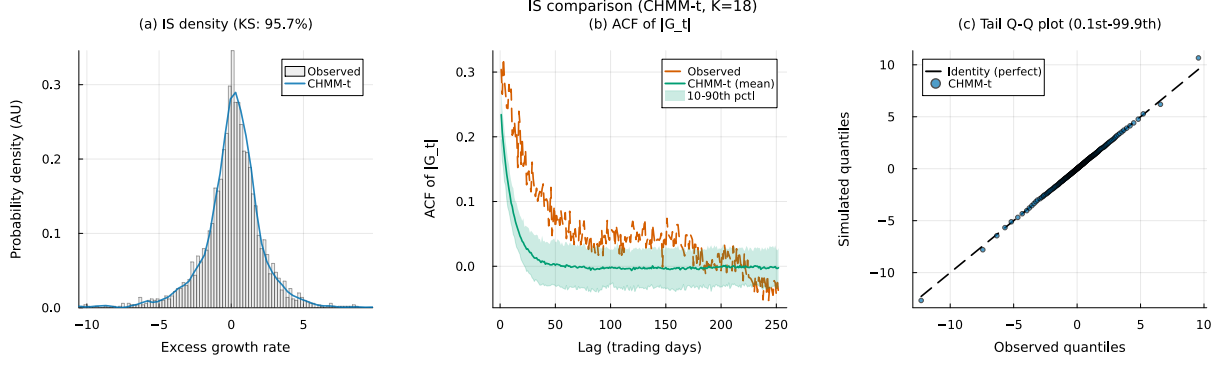


Figure 24. IS SPY model comparison at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) marginal density with KS pass rate, (b) ACF of $|G_t|$, and (c) tail Q-Q plot. The CHMM-N counterpart is Figure 3; the CHMM-L counterpart is Figure 25.

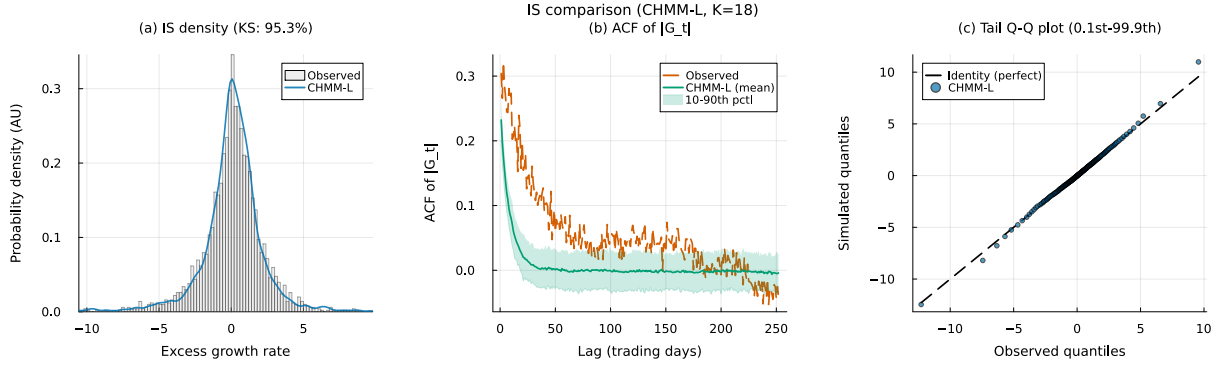


Figure 25. IS SPY model comparison at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 24.

OoS validation. Figures 26–27 show the OoS validation panels for CHMM-t and CHMM-L at $K = 18$. The CHMM-t panel demonstrates the OoS kurtosis alignment noted in the main text (simulated 6.40 vs. observed 6.24), and the CHMM-L panel shows the tighter density fan produced by the heavier Laplace tails relative to the Gaussian.

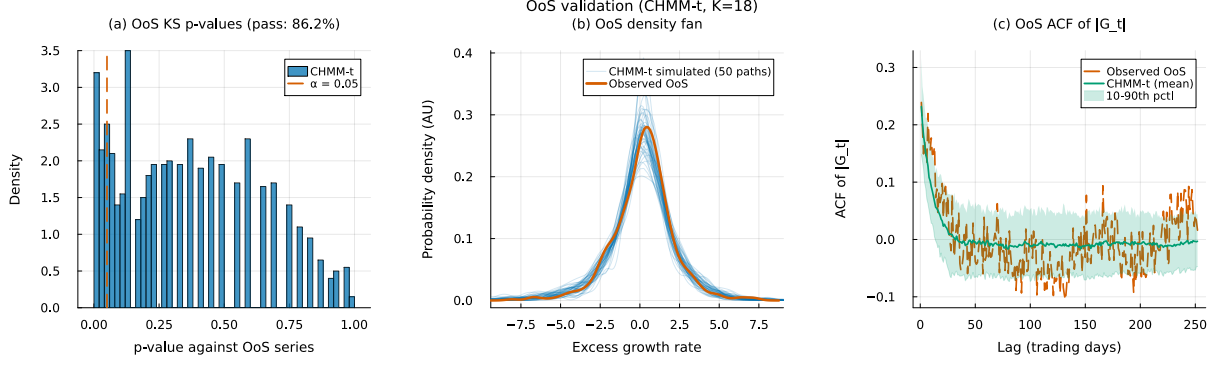


Figure 26. OoS SPY validation at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) KS p -values, (b) OoS density fan, and (c) OoS ACF of $|G_t|$. The CHMM-N counterpart is Figure 4; the CHMM-L counterpart is Figure 27.

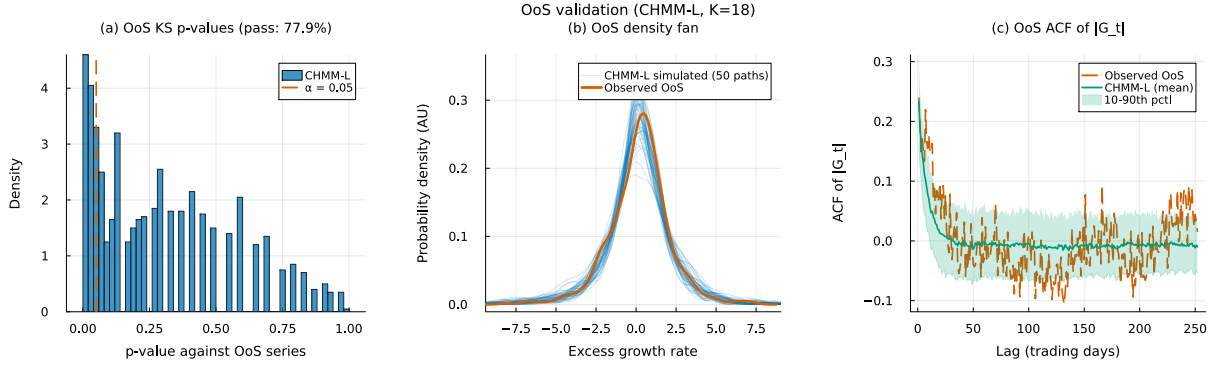


Figure 27. OoS SPY validation at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 26.

Convergence. Figure 28 shows the Baum-Welch / ECM / weighted-EM log-likelihood histories at $K = 18$ for all three families. The CHMM-t and CHMM-L curves inherit the monotone increase of the classical EM guarantee; the CHMM-t curve has a slightly longer pre-plateau phase because each ECM sub-iteration refines ν_k with a golden-section search.

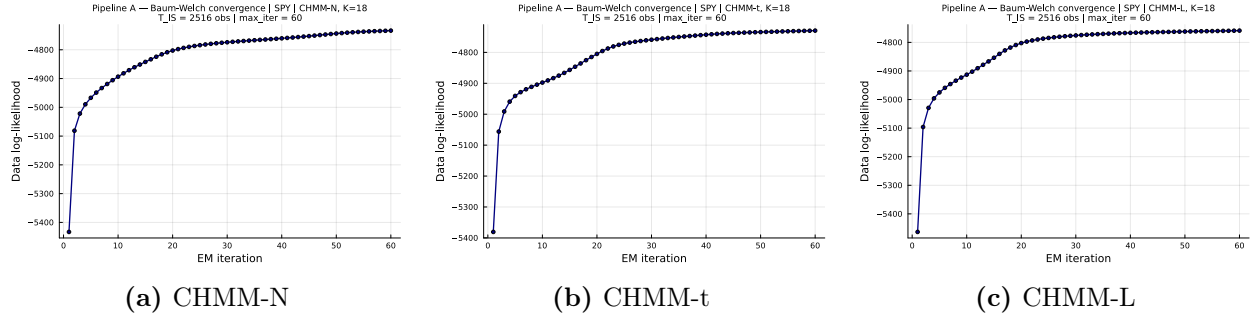


Figure 28. Log-likelihood histories for the SPY IS fit (Pipeline A) at $K = 18$ across the three emission families (CHMM-N Gaussian classical EM; CHMM-t Student-t ECM with per-state ν_k golden-section sub-step; CHMM-L Laplace weighted-EM). All three curves are monotonically non-decreasing and plateau well before the `max_iter` = 60 budget.

A.23 Rydén $K = 2$ Replication

Table 12 reports the direct replication of Rydén et al.’s 1998 $K = 2$ setting on SPY IS under both initialization strategies, as discussed in Section 8. The quantile-based initialization used in the main body is row 1; five independent random-seed initializations are rows 2–6. Random init here draws per-state (μ_k, σ_k) from a moderate perturbation of the sample moments, which is representative of the random-init practice common in the 1990s HMM literature.

Table 12. Rydén replication: $K = 2$ CHMM-N on SPY IS under quantile vs random initialization (seed sub-index varies the random init).

Init	KS IS (%)	AD IS (%)	ACF-MAE	Kurt IS	KS OoS (%)
Quantile (main body)	79.0	71.5	0.0501	2.57	72.4
Random seed = 1	78.7	70.7	0.0502	2.57	73.2
Random seed = 7	75.3	69.6	0.0498	2.56	74.0
Random seed = 42	77.6	70.0	0.0498	2.59	71.8
Random seed = 100	76.2	69.4	0.0500	2.59	72.3
Random seed = 2024	76.4	69.8	0.0499	2.54	72.9

The key observation is that ACF-MAE is essentially constant at ≈ 0.050 across all six rows, matching the $K = 18$ CHMM-N of the main body. IS KS pass rate remains below the $K = 18$ operating point (75.3–79.0%) because two Gaussian components cannot tile the full multi-regime return distribution. The combination is consistent with the reading that the binding low- K constraint in the Rydén et al. [6] setup is distributional rather than temporal: the ACF is already well reproduced at $K = 2$, and it is distributional fidelity that drives the preference for $K = 18$.

A.24 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

A.25 OoS KS Test-Power Calibration

Table 13 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of Section 6.5. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\text{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 13. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\text{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\text{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\text{IS}}, \hat{\sigma}_{\text{IS}}$) at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in Section 6.5 (80–84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.26 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric: (i) the asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and

Table 14. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed 20260422 with $N_{\text{paths}} = 500$. “asyp pass” is the original Section 6.3 metric (two-sample KS p-value ≥ 0.05). “mean pv” is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. “blkL%” is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L . Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383. CSV at `results/robustness/ks_block_bootstrap.csv`.

Generator	asyp pass% $\uparrow$	mean pv $\uparrow$	pv q_{05}	pv q_{95}	blk5% $\uparrow$	blk10% $\uparrow$	blk20% $\uparrow$
i.i.d. bootstrap	99.8	0.812	0.371	0.999	98.2	99.0	99.6
GARCH(1,1)	25.4	0.042	0.000	0.201	5.6	11.2	16.2
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	95.8	0.571	0.059	0.987	82.4	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0

GARCH all carry volatility clustering, so the asymptotic null distribution may not apply; and (ii) the binary “pass / fail at $\alpha = 0.05$ ” reduces a continuous p-value to a one-bit summary. Table 14 reports two recalibrations addressing each concern.

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1,000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and 0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator’s paths are from the observed IS series.

Two patterns emerge that the asymptotic pass-rate metric of the main panel does not surface. First, the cross-generator ordering on the mean p-value scale (bootstrap 0.81, CHMM-t 0.57, CHMM-N 0.55, CHMM-L 0.50, GARCH 0.04) is much more compressed for CHMM-vs-bootstrap than the 94.8–99.8% asyp pass rates suggest: the CHMM family lands at mean p-value ~ 0.5 , the natural value for paths that are statistically indistinguishable from the observed series. Second, GARCH(1,1)’s drop from 25.4% asyp $\rightarrow$ 5.6–16.2% block-aware confirms concern (i): the asymptotic test is mildly lenient on volatility-clustering generators with the wrong marginal, and the block-aware test catches GARCH’s misspecification more sharply. The CHMM family, by contrast, sees only a small block-bootstrap penalty (94.8% $\rightarrow$ 88–92%), indicating that its asymptotic-test passes are not a clustering-induced artefact.

A.27 Bandwidth, Bin-Count, and Multi-Seed Robustness

Fixed observed-sample MMD bandwidth. An auxiliary MMD metric we report in this appendix (Section A.6) computes Maximum Mean Discrepancy with median-heuristic bandwidth on the pooled (observed + simulated) window matrix, so the kernel scale γ varies with the generator under evaluation. We refit MMD across the full extended generator panel under a single bandwidth γ_{obs} computed once on 500 observed standardised 20-day windows from SPY IS. The legacy and fixed MMDs differ only in the third or fourth decimal across all ten generators in our reproduction (full numbers in `results/robustness/mmd_fixed_bandwidth.csv`); the cross-generator ranking under the fixed bandwidth is preserved. The bandwidth-heuristic concern is therefore real in principle but

small in practice for this panel.

Multi-seed Monte Carlo on headline metrics. The paper reports headline statistics under a single global seed. We re-run the CHMM-N / -t / -L pipeline at $K = 18$ across ten alternative global seeds (base 20260422, increment 100) with $N_{\text{paths}} = 250$ per seed and report mean $\pm$ std for the headline metrics in Table 15. Seed-to-seed variability is small: IS KS std is 1.2–1.8%; OoS KS std is 1.8–3.6%; simulated IS kurtosis std is 0.06 for CHMM-N, 0.05 for CHMM-L, and 1.95 for CHMM-t (reflecting the well-known instability of the per-state ν_k ECM at moderate K , already discussed in Section 8). Kupiec $\text{LR}_{\text{uc}}^{1\%}$ has std up to 1.25 for CHMM-N because the breach count moves on a coarse 0–10 grid at $T_{\text{OoS}} = 572$. The single-seed headline numbers in Table 1 (e.g., CHMM-N IS KS 93.8%, simulated kurt 4.99) sit comfortably inside the multi-seed bands. CSV at `results/robustness/multiseed_headline.csv`.

Table 15. Multi-seed Monte Carlo on headline metrics. Ten global seeds; $N_{\text{paths}} = 250$ per seed (reduced from the headline panel’s 1,000 for tractability across 30 fits, so the seed-to-seed std reported here is an upper bound on the std the headline panel would show at $N_{\text{paths}} = 1,000$).

Model	IS KS %	OoS KS %	sim Kurt	ACF-MAE	$\text{LR}_{\text{uc}}^{1\%}$	$\text{LR}_{\text{uc}}^{5\%}$
CHMM-N	94.7 ± 1.2	82.6 ± 1.8	5.03 ± 0.06	0.0467 ± 0.0006	2.12 ± 1.25	3.83 ± 0.00
CHMM-t	94.6 ± 1.2	85.6 ± 1.9	13.95 ± 1.95	0.0518 ± 0.0003	0.54 ± 0.15	3.19 ± 0.34
CHMM-L	94.4 ± 1.8	80.8 ± 3.6	6.71 ± 0.05	0.0537 ± 0.0003	3.26 ± 0.00	2.71 ± 0.59

A.28 Walk-Forward Rolling-Window Re-Estimation (Pipeline A)

This robustness check lives inside Pipeline A: the CHMM-N is still fit per ticker independently, only the fit window rolls forward. It does not introduce cross-asset coupling, and therefore should not be interpreted alongside the Pipeline B results in Section 6.6. Table 16 reports the quarterly walk-forward re-estimation results referenced in Section 6.4. The walk-forward protocol refits the CHMM-N every 63 trading days (one quarter) across the 572-day OoS window, feeding the realized returns of the prior quarter into the training set before each refit. The rolling protocol recovers a modest ~ 1 percentage point on JNJ and a substantial ~ 15 percentage points on JPM relative to the fixed-IS baseline, confirming that the stationarity assumption is the principal contributor to the JPM OoS gap while JNJ’s OoS gap is already small under the fixed fit.

Table 16. Walk-forward (quarterly CHMM-N refit) vs fixed-IS CHMM-N on JNJ and JPM (Pipeline A, per-ticker independent fits; walk-forward only rolls the fit window, it does not introduce cross-asset dependence). $K = 18$, 1,000 simulated paths per (ticker, mode), seed = 20260420. IS window 2014-01-03 through 2024-01-03; OoS window 2024-01-04 through 2026-04-17 ($T_{\text{OoS}} = 572$). The walk-forward mode refits the CHMM-N every 63 trading days across the OoS window, feeding each completed quarter’s realized returns back into the training set before the next refit. Columns report percent KS / AD pass rate at $\alpha = 0.05$, excess kurtosis, ACF-MAE of $|G_t|$, Wasserstein-1, and Hellinger distance.

Mode	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
JNJ fixed-IS	99.3	98.2	94.0	92.1	5.46	4.98	0.0322	0.095	0.0786
JNJ walk-forward	99.3	98.2	94.9	91.3	5.46	5.14	0.0322	0.095	0.0786
JPM fixed-IS	96.6	94.4	49.0	48.6	7.33	6.31	0.0449	0.165	0.0875
JPM walk-forward	96.6	94.4	64.3	63.6	7.33	5.96	0.0449	0.165	0.0875

A.29 Block-Bootstrap Baseline

Table 17 reports the full seven-metric panel for the stationary block bootstrap at block length $b = 10$ trading days, the temporally-aware non-parametric baseline added to Table 1.

Table 17. Stationary block bootstrap (block length $b = 10$ trading days) on SPY (1,000 paths, seed = 20260420).

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	99.2	96.4	7.07	0.0568	0.092	0.0534	100.0
OoS	87.5	86.4	5.34	0.0512	0.225	0.1570	86.9

A.30 GRU Deep-Generative Baseline

Architecture and training details for the GRU baseline used in Section 6.3 are summarised here.

- **Architecture.** A single-layer Gated Recurrent Unit (GRU) with hidden dimension $H = 32$ followed by a Dense head with two outputs $(\mu_t, \log \sigma_t)$. Input is a single feature (the standardised return) over a context window of $w = 20$ trading days.
- **Loss.** Per-window negative log-likelihood of a Gaussian next-step density, $\ell_t = \log \sigma_t + \frac{1}{2}((G_t - \mu_t)/\sigma_t)^2$.
- **Optimiser.** Adam, learning rate 1.0×10^{-3} .
- **Training schedule.** 20 epochs, full pass per epoch over $T_{\text{IS}} - w$ training windows, randomly shuffled at each epoch.
- **Numerical stability.** $\log \sigma_t$ is clamped to $[-6, 4]$ inside the loss to prevent gradient explosion when the network attempts to assign vanishingly-small variance to extreme observations.
- **Pre-processing.** The input return series is standardised by its IS mean and standard deviation; predictions are re-scaled back at sampling time.
- **Synthesis.** The trailing w IS returns seed the recurrence; for each generation step the predicted Gaussian is sampled, the sample is appended to the context window, and the procedure rolls forward. The first 64 steps are discarded as burn-in.
- **Implementation.** Julia using Flux.jl [42] with the Optimisers.Adam state. Random seed for weight initialisation, training shuffling, and per-path sampling: 20260420, with deterministic per-path sub-seeds.

The trained model is callable through the same `simulate_gru(model, n_steps; seed_window)` interface as the CHMM, so the seven-metric panel evaluates the GRU on byte-identical windows to the other generators.

A.31 Per-Ticker OoS Price-Simulation Figures and Full Path-Level Metrics (Unconditional CHMM)

This appendix collects (i) the full per-family path-level probabilistic-forecast metrics for an auxiliary OoS price-simulation diagnostic on the six SPY-member tickers and (ii) the per-ticker price-fan and

Table 18. GRU deep-generative baseline on SPY (single-layer, hidden $H = 32$, window $w = 20$, 20 epochs, Adam $\eta = 10^{-3}$, 1,000 paths, seed = 20260420). Final training NLL: 0.2005.

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	18.1	13.7	2.85	0.0518	0.196	0.0966	37.4
OoS	52.5	55.0	2.21	0.0500	0.293	0.1603	69.7

terminal-distribution figures for NVDA, JNJ, JPM, AAPL, and QQQ. Plotting conventions: blue shaded bands are the 5–95% and 25–75% quantile envelopes of the 500 simulated paths, blue solid line is the simulated median, red solid line is the observed OoS price track. Terminal-histogram panels show the cross-path distribution of $\tilde{P}_T$ at T_{OoS} in gray, with the observed terminal P_T^{obs} as a red vertical line and the simulated median as a blue dashed line. For brevity, the figures show CHMM-N only (the representative family; differences across the three emission families in the price-level metrics are small, see Table 19).

Table 19. Full path-level probabilistic-forecast metrics for OoS equity-price simulation, all (ticker, family) cells ($K = 18$, 500 paths, $T_{\text{OoS}} = 572$; global seed policy in Section 3.5). $\overline{C}_{0.90}$: fraction of OoS steps at which the observed price falls inside the simulated 5–95% band (calibration); $\text{MAPE}_{0.50}$: mean absolute percentage error between observed path and simulated median (sharpness); $\overline{\text{CRPS}}$: horizon-average bias-corrected sample CRPS of Gneiting and Raftery [43]; $\overline{\text{CRPS}}/S_0$: CRPS scaled by S_0 to remove price-level effects. Lower MAPE and CRPS, higher $\overline{C}_{0.90}$, are better. **Headline:** no CHMM emission family dominates on all four metrics across all six tickers; differences between CHMM-N, CHMM-t and CHMM-L live mostly in the return tails and propagate only weakly into aggregated price-level scores.

Ticker	Family	$\overline{C}_{0.90}$	$\text{MAPE}_{0.50}$	$\overline{\text{CRPS}}$	$\overline{\text{CRPS}}/S_0$
SPY	CHMM-N	1.000	0.100	39.25	0.0835
	CHMM-t	1.000	0.112	42.66	0.0908
	CHMM-L	1.000	0.102	38.68	0.0823
NVDA	CHMM-N	0.698	0.339	30.62	0.6424
	CHMM-t	0.636	0.383	34.77	0.7293
	CHMM-L	0.666	0.411	36.58	0.7674
JNJ	CHMM-N	1.000	0.101	12.55	0.0780
	CHMM-t	1.000	0.098	12.62	0.0784
	CHMM-L	1.000	0.101	12.65	0.0786
JPM	CHMM-N	1.000	0.195	32.23	0.1881
	CHMM-t	1.000	0.241	40.38	0.2357
	CHMM-L	1.000	0.211	35.25	0.2057
AAPL	CHMM-N	1.000	0.108	21.30	0.1156
	CHMM-t	1.000	0.075	18.90	0.1025
	CHMM-L	1.000	0.077	18.82	0.1021
QQQ	CHMM-N	0.998	0.073	30.48	0.0763
	CHMM-t	0.995	0.079	31.63	0.0792
	CHMM-L	1.000	0.118	41.12	0.1029

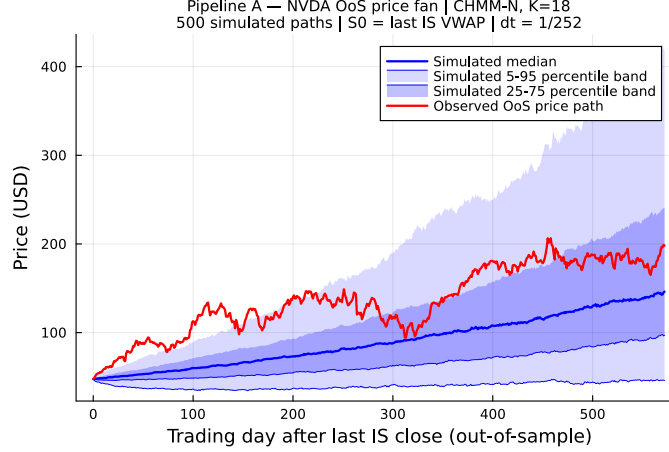


Figure 29. NVDA OoS price fan, CHMM-N ($S_0 = 47.67 \rightarrow P_T^{\text{obs}} = 198.07$, a $4.15\times$ rally). The observed path exits the 25–75% interquartile band by mid-horizon but remains inside the 5–95% envelope; horizon-wide coverage under CHMM-N is 0.70 (CHMM-t: 0.64, CHMM-L: 0.67; Table 19), the lone calibration outlier in the panel.

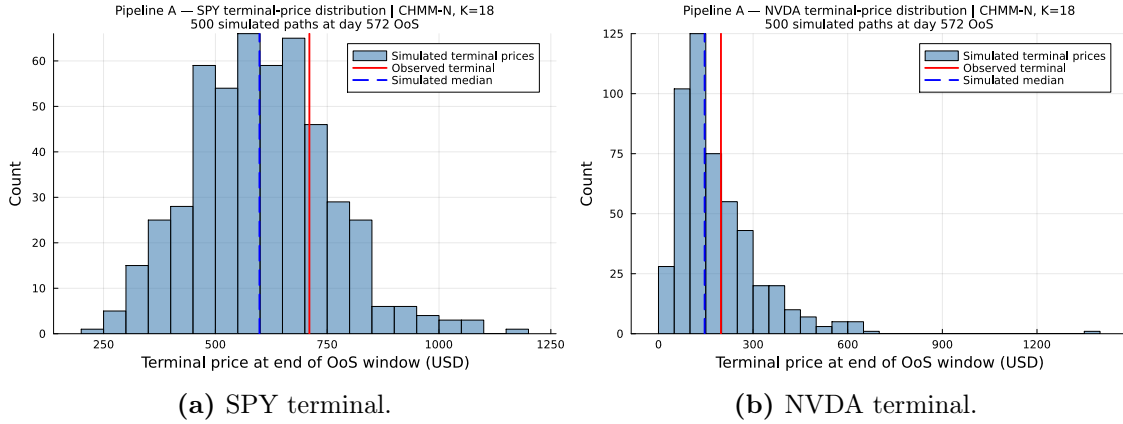


Figure 30. OoS terminal-price distributions under CHMM-N for SPY and NVDA. The right-skew is characteristic of log-returns aggregated to prices; the SPY terminal sits close to the simulated median, while the NVDA terminal sits in the upper-right of the simulated distribution (the realised rally is in the upper tail of the CHMM-N terminal density).

A.32 Expanded GARCH-Family Baselines

We add five conditional-volatility baselines to the single GARCH(1,1) Gaussian row of Table 1. All are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window ($T_{\text{IS}} = 2,516$); simulation follows the same recursion used in estimation. Model specifications are below.

EGARCH(1,1) Gaussian. The exponential GARCH of Nelson [44] evolves $\log \sigma_t^2$ in levels:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha \left(|z_{t-1}| - \sqrt{2/\pi} \right) + \gamma z_{t-1}, \quad z_t = (r_t - \mu)/\sigma_t, \quad z_t \sim \mathcal{N}(0, 1). \quad (23)$$

The log-variance formulation removes the GARCH positivity constraints, and the leverage term γz_{t-1} captures the asymmetric response to negative returns.

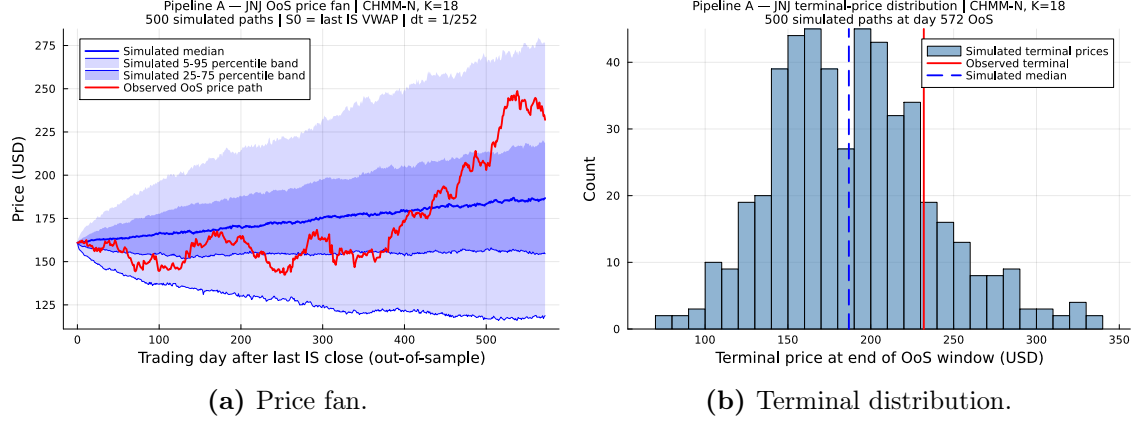


Figure 31. JNJ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as above).

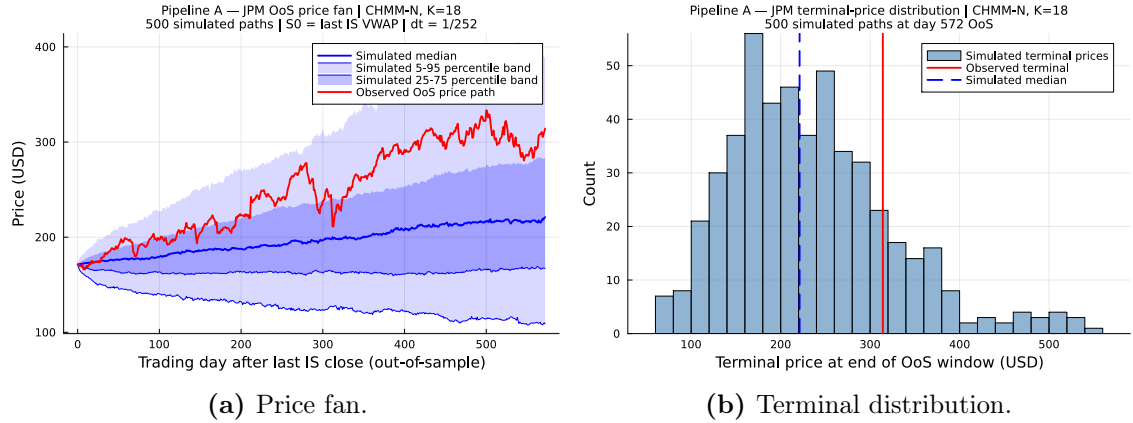


Figure 32. JPM OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as above).

GJR-GARCH(1,1) Gaussian. The threshold GARCH of Glosten et al. [45] / Zakoian [46] augments GARCH(1,1) with an indicator on negative innovations:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}\{\varepsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2, \quad (24)$$

with $\varepsilon_t = r_t - \mu$. Stationarity requires $\alpha + \beta + \gamma/2 < 1$; positive γ indicates leverage.

GARCH(1,1) Student-t innovations. A direct tail-aware competitor to CHMM-t: the GARCH(1,1) variance recursion is unchanged, but innovations follow a standardised Student-t with $\nu > 2$ degrees of freedom rescaled to unit variance. The log-likelihood uses the Student-t density:

$$r_t - \mu = \sigma_t \cdot z_t, \quad z_t = \eta_t \sqrt{(\nu - 2)/\nu}, \quad \eta_t \sim t_\nu. \quad (25)$$

The fitted $\hat{\nu} = 6.89$ on SPY IS sits inside the moderate-tail regime the equity-risk literature typically reports.

HAR-RV on daily squared returns. The heterogeneous-autoregressive model of Corsi [47] models realised variance as a mixture of daily, weekly, and monthly components:

$$\log RV_t = \beta_0 + \beta_d \log RV_{t-1} + \beta_w \log \overline{RV}_{t-5:t-1} + \beta_m \log \overline{RV}_{t-22:t-1} + \eta_t. \quad (26)$$

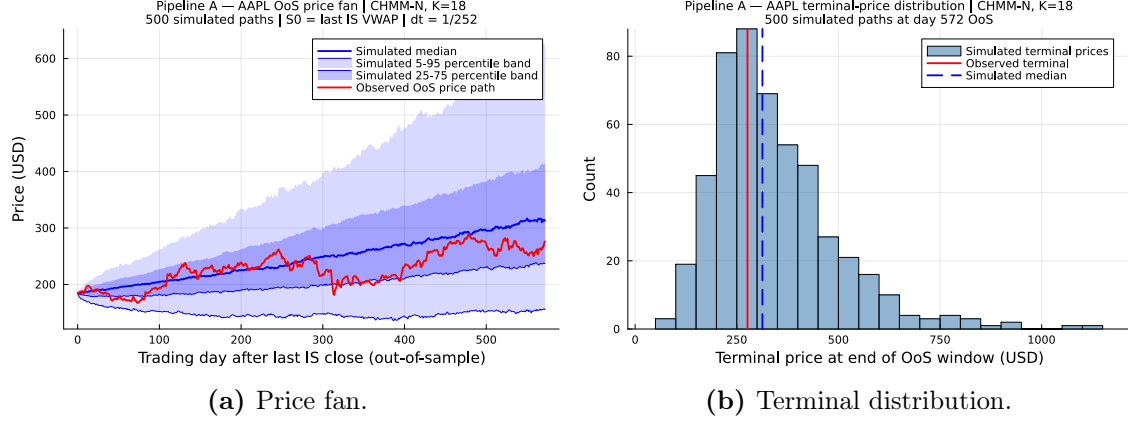


Figure 33. AAPL OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as above).

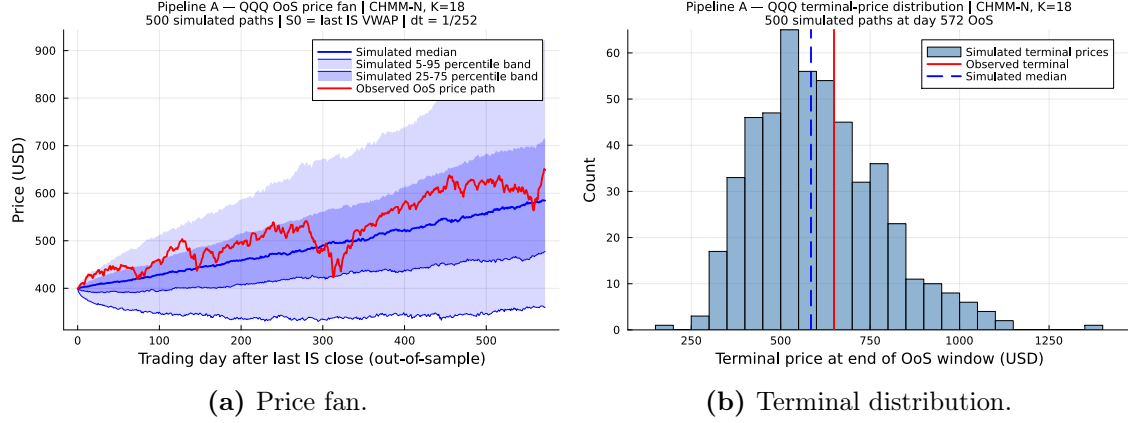


Figure 34. QQQ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as above).

In the absence of intraday bars we use daily squared demeaned returns as the RV proxy; simulation samples $\eta_t \sim \mathcal{N}(0, \hat{\sigma}_\eta^2)$ and rolls forward the recursion, then draws $r_t - \mu = \sqrt{RV_t} z_t$ with $z_t \sim \mathcal{N}(0, 1)$.

MS-GARCH $K = 3$. A three-regime extension of the Markov-switching GARCH(1,1) ($G_t = \mu + \sigma_t z_t$ with regime-specific recursion $\sigma_t^2(k) = \omega_k + \alpha_k(G_{t-1} - \mu)^2 + \beta_k \sigma_{t-1}^2(k)$). Parameter layout is three regime triples $(\omega_k, \alpha_k, \beta_k)$ plus six transition-matrix logits (softmax-normalised per row, enforcing a non-trivial diagonal); fit by grid-initialised Nelder-Mead. States are canonicalised by ascending unconditional variance, producing the calm / medium / stress decomposition $\sigma = [0.10, 1.77, 10.04]$ on SPY IS.

A.33 Cross-Asset Dependence: Full Pipeline B Panel

A.34 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in Section 6.6 of the main text. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A (Section 6.4) and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar’s theorem and the rank-reordering simulator. Sklar’s theorem [20, 21] states that any d -dimensional joint CDF $H(g_1, \dots, g_d)$ with continuous marginals $F_1, \dots, F_d$ can be written as $H = C(F_1, \dots, F_d)$ for a unique copula C . The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j -th column is an i.i.d. sample from F_j , and $\mathbf{U}$ is an independent sample from C , then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$ ’s produces a sample from H , up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [24].

Kendall’s τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall’s τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [23, 28]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν . The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0, 1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \dots, x_d)^\top$, is

$$\begin{aligned} \log c_t(\mathbf{u}; \Sigma, \nu) &= \log \Gamma\left(\frac{\nu+d}{2}\right) + (d-1) \log \Gamma\left(\frac{\nu}{2}\right) - d \log \Gamma\left(\frac{\nu+1}{2}\right) - \frac{1}{2} \log |\Sigma| \\ &\quad - \frac{\nu+d}{2} \log \left(1 + \frac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^d \frac{\nu+1}{2} \log \left(1 + \frac{x_j^2}{\nu}\right). \end{aligned} \quad (27)$$

Summing over $t = 1, \dots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall’s- τ estimate. We evaluate equation (27) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 6.6 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d -dimensional Gaussian copula with correlation Σ , we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν , we draw $W \sim \chi_\nu^2/\nu$ independent of Y and form the multivariate- t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 20 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 6.6. QQQ’s high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ’s lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 21.

Table 20. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\text{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2,516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	-0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.35 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 35 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in Section 6.6. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature.

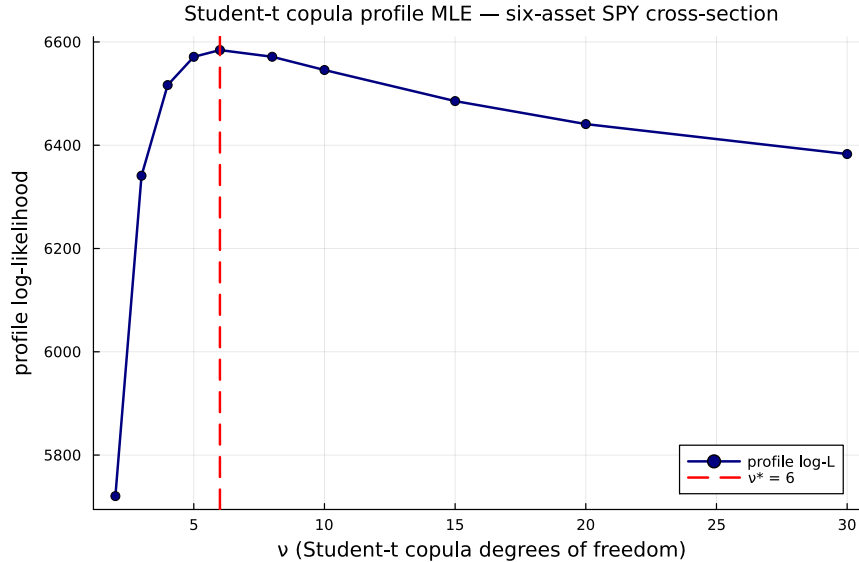


Figure 35. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

A.36 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 21)

Figure 36 visualises the observed and path-averaged simulated correlation matrices for the dependence models of Section 6.6. The panel ordering is SIM, Gaussian copula, Student-t copula (each reported quantitatively in Table 21), plus a truncated C-vine variant with SPY as root for visual comparison.

All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.

Pipeline B — Cross-asset correlation reproduction | $K=18$ marginals, 200 paths, IS window | top-left is the data

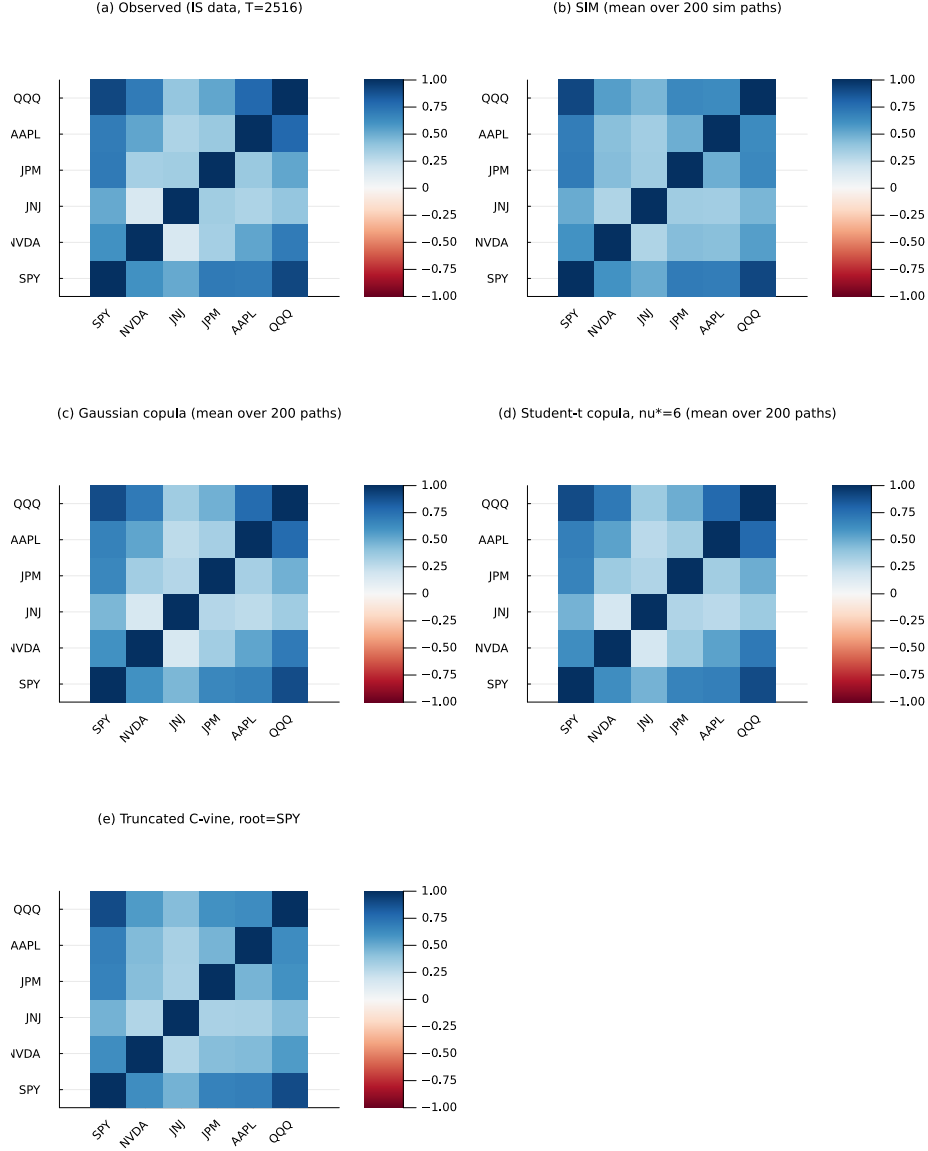


Figure 36. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 21). Panel (a), top-left: *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{IS} = 2,516$). Panel (b), top-right: path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c), middle-left: path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d), middle-right: path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e), bottom-left: path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS} . All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 3; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.37 Per-Asset KS Bar Chart (Pipeline B, Sanity Check for Table 21)

Visualisation of the per-asset KS pass rates from Table 21; the SIM-vs-copula gap on JPM and AAPL is the visual headline.

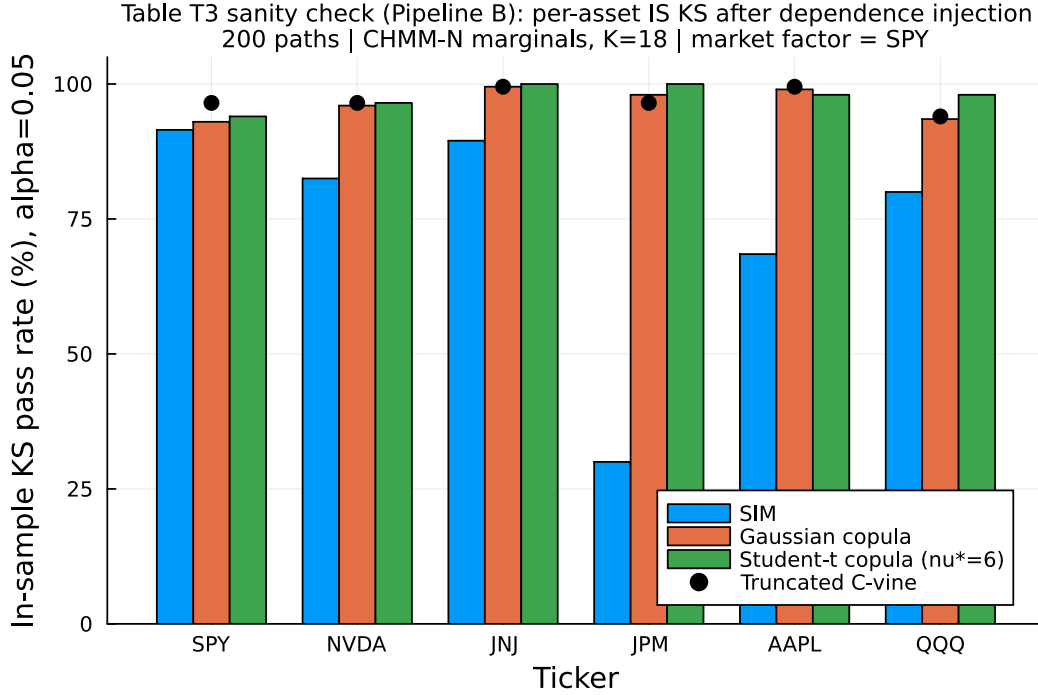


Figure 37. Per-asset IS KS pass rates after dependence injection (Pipeline B sanity check for Table 21). For each of the six tickers and each of the three dependence constructions (SIM, Gaussian copula, Student-t copula with $\nu^* = 6$), the bar reports the percentage of 200 simulated paths whose marginal distribution passes the two-sample KS test against the observed IS series at $\alpha = 0.05$. The market-factor asset SPY is approximately tied across constructions (it is the SIM engine). For non-market assets the copulas dominate because rank reordering preserves each CHMM marginal exactly; the SIM’s affine propagation visibly degrades JPM and AAPL.

A.38 Pre-OoS Validation K -Selection, Walk-Forward Summary, and ν_k Diagnostics

Pre-OoS validation K -selection. The $K = 18$ operating point in the main panel was chosen via a multi-objective criterion combining IC rank with IS and OoS distributional pass rates, which uses the same 2024–2026 window later used for VaR. As a clean re-selection that decouples the two, we fit CHMM-N on 2014-01-03 through 2021-12-31 ($n = 2,013$) and score every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03 ($n = 502$); the 2024–2026 window is not touched. Held-out log-lik, BIC, HQC, and CAIC all select $K^* = 3$ (per-observation log-lik -2.5345 at $K = 3$ vs. -2.5694 at $K = 18$); AIC selects $K^* = 6$; held-out two-sample KS selects $K^* = 9$. None of the six criteria that avoid the OoS window select $K = 18$. The $K = 18$ choice is best read as a multi-metric one that weights tail fidelity (kurtosis, AD) and downstream VaR behaviour alongside likelihood; a parallel main panel at $K = 3$ would re-rank some rows in CHMM’s favour on ACF-MAE and against it on tail fidelity. The 2022–2023 validation slice is itself partly a structural-break period (Section 6.5), so any model with substantial IS-specific structure generalises worse on it.

ν_k bracket and penalised-ECM rate sweep. A bracket sweep $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0\}$ shows the CHMM-t fit does not pile against the lower bracket: median $\nu_k = 50$, only 2/18 states near the lower edge. Lifting $\nu_{\min}$ does not reduce the $14.30 \rightarrow 7.69$ overshoot because the fit shifts posterior mass into the newly-recentred tail states. An exponential $1/\nu_k$ shrinkage prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduces simulated excess kurtosis monotonically: 14.30, 11.19, 8.43, 5.41, 3.96, 3.67. The recommended rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 (close to the observed 7.69) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moves ($2.1 \rightarrow 4.89$); upper bracket and median remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Cross-asset summary. The full per-construction Pipeline B panel is in Section A.33; for reference, the headline ordering is summarised in Table 21.

Table 21. Cross-asset dependence summary (Pipeline B; CHMM-N marginals at $K = 18$; 200 paths). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. The Student-t copula at $\nu^* = 6$ is the strongest dependence layer on every IS metric; full per-asset detail is in Section A.33.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	78.5	85.0	0.076	0.261
Gaussian copula	97.8	87.2	0.031	0.202
Student-t copula ($\nu^* = 6$)	96.0	89.8	0.027	0.208